$p + p \to \pi^+\pi^-e^+e^-$ Analysis Report

HADES, p+p at $T = 4.5\,\mathrm{GeV}$ (feb22 beam time)
Part I: the simulation basis

W. Przygoda

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Status of this draft. Part I uses simulation only. Two inputs are still provisional and are kept as parameters: (i) the integrated luminosity of the data, $L_{\mathrm{data}} = 6080.5\,\mathrm{nb}^{-1}$, taken from the producers' note and not yet confirmed by the luminosity analysis; (ii) the physics weights of the simulated η decays (Sect. 3.13). The generator's $\eta \to \pi^+\pi^-e^+e^-$ has a non-QED $M(e^+e^-)$ distribution; it is now corrected by weights (main variant V3), as are the Dalitz plot of $\eta \to \pi^+\pi^-\pi^0$ and the pion form factor of $\eta \to \pi^+\pi^-\gamma$. Numbers marked "(pre-rw)" are before these weights.

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Table 1: Notation and abbreviations. Last column: where the item is defined in detail.

Symbol	Meaning	Details
Programs and samples		
SMASH, PLUTO	event generator of p+p collisions; decay generator used by the converter	§2
converter	`convert-smash`: SMASH output → HGeant input, forces decays in the lepton and η sets	Tab. 2
HGeant, DST	GEANT3-based HADES detector simulation; reconstructed data	Tab. 4
PAT, FAT	PostDST Analysis Tool (hypotheses, PID, relative time, ranking); Final Analysis Tool (selection, kinematics, weights)	§4
std, lep, η set	simulation sets: natural decays; every π^0, η, ρ^0, ω, ϕ forced into dileptons; every η forced into $\pi^+\pi^-e^+e^-$ or $\pi^+\pi^-$ X17	Tab. 2
N, $b_{\max}$, T	events per set (10^{10}); largest impact parameter (1.2 fm); beam kinetic energy	Tab. 3
L_{sim}, L_{data}	luminosity of one simulated set (221.05 nb^{-1}); of the data (6080.5 nb^{-1}, not confirmed)	Tab. 3
row	one combination of tracks of a hypothesis; an event can give several rows	§4
event group	the adjacent rows of one event (used for the statistical errors)	§3
isBest	PAT ranking of the combinations of an event: 1 best, 0 others; "all combinations" = isBest ≥ 0	§4
PT3, `trigbit`	the physics trigger of this analysis; trigger word (PT3 = bit 8192; simulation 12288)	§3
"pre"	base selection: isBest ≥ 0, vertex $z > -500$ mm, PT3, vertex $z > -160$ mm, corrected momenta	§4.3
Generator truth, classes and weights		
`geninfo`, `geninfo1`, `geninfo2`	generator codes of a GEANT track: decay code, code of the direct parent, counter of the decay in the event	§3.1, §3.2
forced decay	a decay chosen by the converter instead of the natural one, with a stored weight w	Tab. 7
w, W	generator weight stored on a track; physical weight of a row	§2.4
rules v1, v2	weight rules; v2 uses the exact HGeant branching ratios of the card `ETA`	§3.3

Symbol	Meaning	Details
mode A, B	ownership of row types between the sets (A default)	§2.4
classes 0–8	four-particle truth class (`sim_cls`); signal classes 2–6	Tab. 10
L0–L11, P0–P9	lepton-pair and pion-pair truth classes	Tab. 10
T1–T4	templates: signal, physical background, fake lepton, leptons from two sources	Tab. 10
G_η, $G_{\pi^0\mathrm{D}}$, G_conv, G_vec, G_oth	groups of T2 by the source of the pair and the pions; multi-pion = $G_{\pi^0\mathrm{D}} + G_\mathrm{conv} + G_\mathrm{vec}$	Tab. 20
ghost	reconstructed track flagged as not matching one GEANT particle (`sim_isghost`)	Tab. 10
pre-rw, V0–V4	before the physics reweighting of the η decays; its model variants (V3 main)	§3.13
Cross sections and statistics		
σ_prod	production cross section in SMASH: mean number per collision $\times\, \pi b_\mathrm{max}^2$	Tab. 9
BR	branching ratio (PDG unless stated)	Tab. 7, 8
ε, N_gen	rows after "pre" per generated decay; number of generated decays	Tab. 12
σ_vis, σ	visible cross section $\sum W/L_\mathrm{sim}$ of rows; $= \sigma_\mathrm{prod} \times \mathrm{BR} \times \varepsilon$	§2.5
N_eff	effective number of rows, $(\sum W)^2/\mathrm{Var}$	§5
CB, k	combinatorial background of the data, $2\sqrt{N_{++}N_{--}}$ times k (absent in simulation)	§4
Particle identification		
β_new	velocity from the relative times of flight of the tracks of a hypothesis	§4.2
charge × momentum	horizontal axis of the PID plots (negative for negative tracks)	Fig. 1
$n\sigma$ cut, v1, v6	PID polygon at n standard deviations; experiment (v1) and simulation (v6) cut files	§4.2
ring, `kIsLepton`	RICH ring correlated with the track; PAT puts such tracks into lepton slots	§4.2
Kinematics and cuts		
$M(\pi^+\pi^-e^+e^-)$, $M(e^+e^-)$, $M(\pi^+\pi^-)$	invariant masses of $\pi^+\pi^-e^+e^-$, of the e^+e^- pair, of the $\pi^+\pi^-$ pair	
MM_4	missing mass of $\pi^+\pi^-e^+e^-$: mass of (beam + target − $\pi^+\pi^-e^+e^-$) four-momenta; for $pp \to pp\eta$ it is $M(pp)$	Tab. 18
$p_{T,4}$, y_4	transverse momentum (lab) and rapidity (beam–target c.m.) of $\pi^+\pi^-e^+e^-$	Tab. 18
OA	opening angle between e^+ and e^- (lab)	Tab. 18
α_lab	angle between the $\pi^+\pi^-$ and e^+e^- momenta (lab)	Tab. 18
α_η	angle between $\pi^+\pi^-$ and e^+e^- in the rest frame of a hypothetical η (measured $\pi^+\pi^-e^+e^-$ momentum, mass m_η)	Tab. 18
Δz_π	largest $\lvert z_\mathrm{track} - z_\mathrm{vertex}\rvert$ of the two pions	Tab. 18
pions in vertex fit	both pion tracks used in the event-vertex fit (`isUsedVertex`)	Tab. 18
named selections	pre, pre+OA> 2, slides, old code, peak nominal, bump nominal	Tab. 19
bump, peak	$\eta \to \pi^+\pi^-\pi^0$ seen below m_η; η at m_η (direct decay or conversion)	§1
peak / bump window	$0.5333 \le M(\pi^+\pi^-e^+e^-) < 0.5587$ / $0.4576 \le M(\pi^+\pi^-e^+e^-) < 0.5255\,\mathrm{GeV}/c^2$ (simulation)	Tab. 20
Optimisation and blind test (Section 5)		
S, B, P	signal rows, other T1+T2 rows of a window, purity $S/(S+B)$	§5.1
F, F_best	figure of merit $S/\sqrt{S+B}$; its maximum over the scanned cuts	§5.5
PEAK, DIRECT, BUMP	optimisation targets: classes 4+5, class 5, classes 2+3 in their window	§5.5
plateau, freeze rule	cuts with $F \ge 0.95\,F_\mathrm{best}$; the rule that picks the frozen cut	§5.5
half A, B	simulated events split by event-group number (even/odd): templates / pseudo-data	§5.7
toy, μ	one pseudo-data set; the expected (true) yield of half B	§5.7
bias, pull, coverage	mean (extracted $-\mu$); (extracted $-\mu$)/quoted error; fraction of toys with $\lvert Y-\mu\rvert \le$ error	Tab. 31
M1–M3c, B1–B4, D5	extraction methods for the peak, the bump, the direct decay	§5.7
X1–X5	X17 estimators	§5.7
UL, $\mathrm{UL}_\mathrm{stat}$, s_1	expected 90 % CL limit with / without the simulation error of B; X17 rows per unit BR	Eq. (5)

1 Scope

The analysis looks for the η meson in the final state $\pi^+\pi^-e^+e^-$ of p+p collisions at $T = 4.5\,\text{GeV}$. Two structures are expected in the invariant mass $M(\pi^+\pi^-e^+e^-)$:

- the **peak** at m_η from the decay $\eta \to \pi^+\pi^-e^+e^-$, and from $\eta \to \pi^+\pi^-\gamma$ when the photon converts into an e^+e^- pair in the detector;
- the **bump** below m_η from $\eta \to \pi^+\pi^-\pi^0$ followed by the Dalitz decay $\pi^0 \to e^+e^-\gamma$ (or by a photon conversion), when the photon is not detected.

The bump is much larger than the peak and has a known branching ratio, so it can serve as a reference. The e^+e^- pairs of the peak are also the place to search for a hypothetical X17 boson ($\eta \to \pi^+\pi^-X17$, $X17 \to e^+e^-$).

This first part documents the simulation used for the analysis: the samples, their statistics and normalisation, the weights, several technical pitfalls met on the way, and a short description of the selection. The data analysis and the comparisons follow in later parts (Sect. 6).

2 Simulation samples

2.1 Event generator and the three sets

All simulated events come from one SMASH run: p+p at $T = 4.5\,\text{GeV}$, SMASH-3.0, collider mode, impact parameter sampled uniformly in the area in $[0, 1.2]\,\text{fm}$. Every sampled event is kept, including the 4.3 % without any interaction. The converter (`convert-smash`) turns the SMASH output into HGeant input; it was run three times on the *same* 10^{10} events, with different treatment of some particles (Table 2). This was verified: the converted files of the three sets are identical event by event apart from the forced particles, and the reconstructed rows of different sets contain the same generator pions (Sect. 3.4). The three sets therefore describe the same collisions; only the detector simulation (HGeant) is independent in each set.

Table 2: The three SMASH simulation sets. "Forced" means that the converter (with PLUTO) replaced the natural decay of every such particle by the listed channel and stored a weight on its products. Weight 0 in the file means "not weighted" (natural probability).

Set	Converter run	What is changed	Weight stored on the products
standard (`std`)	2026-05-18, `-u 0`, no PLUTO	nothing; π^0 and η decay in HGeant; SMASH resonance decays (ρ, ω, ϕ) are only flagged in the generator code	0 for every particle
lepton (`lep`)	2026-02-11, PLUTO, `database_decays_leptons.dat`	every π^0, η, ρ^0, ω, ϕ forced into a dilepton channel (Table 7)	$w = n_\text{ch} \times \text{BR}$ on the forced-decay products
η (`eta`)	2026-06-03, PLUTO, `database_decays_eta.dat`	every η forced: 50 % $\eta \to \pi^+\pi^-e^+e^-$, 50 % $\eta \to X17\,\pi^+\pi^-$, $X17 \to e^+e^-$ ($m_{X17} = 17\,\text{MeV}/c^2$)	$w = 2$ on the four products

Table 3: Normalisation, common to the three sets.

Events per set	$N = 10^{10}$ (100 000 files $\times$ 100 000 events)
Area per event	$\pi b_{\max}^2/N = 45.239\,\text{mb}/10^{10} = 4.5239 \times 10^{-3}\,\text{nb}$
Integrated luminosity per set	$L_{\text{sim}} = N/(\pi b_{\max}^2) = 221.05\,\text{nb}^{-1}$
SMASH cross sections (for orientation)	$\sigma_{\text{tot}} = 43.29 \pm 0.03\,\text{mb}$, $\sigma_{\text{el}} = 12.72 \pm 0.06\,\text{mb}$, $\sigma_{\text{inel}} = 30.57 \pm 0.07\,\text{mb}$
Expected rows in data	$N_{\text{data}} = (L_{\text{data}}/L_{\text{sim}}) \sum W_{\text{row}}$; for $L_{\text{data}} = 6080.5\,\text{nb}^{-1}$: 27.51 per unit weight

Because $\sigma_{\text{tot}} < \pi b_{\max}^2$, the range of b covers every SMASH collision. The luminosity of a set does not depend on the SMASH cross sections; the particle yields per collision do.

2.2 Detector simulation and PAT production

The three sets went through HGeant and the HADES reconstruction (DST production feb22, `gen4`). The DSTs were processed with PAT (PostDST Analysis Tool) into the three hypothesis trees of this analysis. Table 4 lists the production.

Table 4: PAT production of the simulation (all three sets with the same frozen binary).

PAT branch, commit	`pp45_sim_pippimepem`, `e674942` (binary built 2026-11-11 20:42)
Software	Hydra 2.7.6, ROOT 6.36.02 (Debian 12)
PID cut file	`pp45_pid_cuts_sim_betanew_v6.root`: p, π^+, π^- at 3σ, $e^\pm$ at 5σ (Sect. 4.2)
Hypotheses written	`PipPimEpEm_ID`, `PipPimEpEp_ID`, `PipPimEmEm_ID` (level after the PID cut)
DST input	`/lustre/hades/dstsim/feb22/p4500p/gen4/smash/smash_<set>/root/`, 100 000 files per set, all processed, 10^{10} events per set
Merged files	`ppee_std_all.root` (0.30 GB), `ppee_lep_all.root` (10.0 GB), `ppee_eta_all.root` (4.0 GB)

Table 5: Rows (combinations) and events of the PAT hypothesis trees. "Events" counts the events with at least one row.

	`PipPimEpEm_ID`		`PipPimEpEp_ID`		`PipPimEmEm_ID`	
Set	rows	events	rows	events	rows	events
std	238 074	220 532	1 146	1 066	4 850	4 455
lep	7 622 819	7 035 072	207 380	197 954	498 807	469 069
eta	3 508 574	2 869 528	14 042	12 391	54 156	45 879

2.3 Analysis-framework processing and statistics

The PAT trees are read by FAT (Final Analysis Tool, `main.cc`), which applies the base selection of Sect. 4.3, computes the kinematics and the truth classes, and writes one row per combination into the ntuple `scan_nt`. Table 6 gives the statistics after each step.

Table 6: Statistics after the analysis framework. "FAT rows": isBest ≥ 0, vertex $z > -500\,\text{mm}$, PT3 trigger bit. "Owned": rows that this set provides in the partition of Sect. 2.4 (mode A). $\sum W$: sum of the physical weights of the owned rows with $z > -160\,\text{mm}$; $\sigma = \sum W / L_{\text{sim}}$ is the visible cross section (Sect. 2.5) for producing such a row (pre-rw for the η set).

Set	Channel	PAT rows	FAT rows	$z > -160$	owned	σ [nb]
std	$\pi^+\pi^-e^+e^-$	238 074	230 469	230 115	115 414	506.73
	$\pi^+\pi^-e^+e^+$	1 146	1 133	1 132	1 055	3.911
	$\pi^+\pi^-e^-e^-$	4 850	4 791	4 777	4 599	18.85
lep	$\pi^+\pi^-e^+e^-$	7 622 819	7 441 490	7 432 170	7 082 339	329.36
	$\pi^+\pi^-e^+e^+$	207 380	205 527	205 288	159 543	0.285
	$\pi^+\pi^-e^-e^-$	498 807	495 711	495 101	328 641	0.608
eta	$\pi^+\pi^-e^+e^-$	3 508 574	3 366 025	3 357 676	3 180 398	3.290
	$\pi^+\pi^-e^+e^+$	14 042	13 881	13 856	13 061	0.0134
	$\pi^+\pi^-e^-e^-$	54 156	53 628	53 510	49 892	0.0586

The large row counts of the lepton and η sets do not mean large cross sections: each of their rows carries a small weight. For example, the 7.08 million owned $\pi^+\pi^-e^+e^-$ rows of the lepton set add up to $\sum W = 72\,804$, the 115 414 rows of the standard set to 112 014. The statistical power of a weighted sample is set by its effective number of rows, $(\sum W)^2 / \sum W^2$, not by its row count.

2.4 Weights

Every row gets a physical weight W, so that $\sum W / L_{\text{sim}}$ is a cross section and the three sets can be added. The rules (version 2, `src/sim_truth.h`) are:

1. **Standard set.** $W = 1$, with corrections of the HGeant branching ratios to PDG values for decays that HGeant performs (Table 8). The largest correction is for $\eta \to e^+e^-\gamma$: the HGeant configuration of this production sets its branching ratio to 8.8 times the PDG value (Sect. 3.3), so an e^+e^- pair from an η Dalitz decay gets the factor 0.114.

2. **Lepton set.** W is the product of the weights of the *distinct* forced decays found among the four tracks (Table 7), with the PDG/database branching-ratio correction. A track from no forced decay contributes the factor 1. Two tracks belong to the same forced decay when their codes (`geninfo`, `geninfo2`) agree (Sect. 3.2).

3. **η set.** A row with the four products of one forced decay gets $W = 2 \times \text{BR}$: $\text{BR}(\eta \to \pi^+\pi^-e^+e^-) = 2.68 \times 10^{-4}$ (PDG), and a hypothesis value for X17 (0 by default). Everything else in this set is the standard set without the natural η decays.

4. **Ownership.** One row type must come from one set only. In mode A (default) the lepton set provides every row with a forced-decay product of a π^0, η, ρ^0, ω or ϕ, and the standard set drops such rows; the η set provides every row with a product of a forced η decay. Mode B (variant) leaves the π^0 and η Dalitz rows in the standard set.

Table 7: Forced decays of the lepton set. The converter picks one of the n_{ch} active channels of a mother with equal probability, so the stored weight is $n_{\mathrm{ch}} \times \mathrm{BR}$.

Decay	n_{ch}	BR (database)	stored w	BR (PDG)	PDG/database
$\pi^0 \to e^+e^-\gamma$	1	1.198×10^{-2}	0.01198	1.174×10^{-2}	0.980
$\eta \to e^+e^-\gamma$	1	0.6×10^{-2}	0.006	0.69×10^{-2}	1.15
$\rho^0 \to e^+e^-$	1	4.48×10^{-5}	4.5×10^{-5}	4.72×10^{-5}	1.054
$\omega \to e^+e^-$	2	7.15×10^{-5}	1.43×10^{-4}	7.38×10^{-5}	1.032
$\omega \to \pi^0e^+e^-$	2	5.9×10^{-4}	1.18×10^{-3}	7.7×10^{-4}	1.305
$\phi \to e^+e^-$	1	2.99×10^{-4}	2.99×10^{-4}	2.979×10^{-4}	0.996

Table 8: Decays of π^0 and η performed by HGeant, and the correction factors applied to rows with such decays (rules v2). "Card": the branching ratios in percent set in the HGeant configuration of this production (π^0: HGeant default, no card); HGeant normalises them to their sum. The producers' measurement on 10^6 events (used in rules v1) agrees with the normalised card within 0.3–2.2 standard deviations. The two $\pi^+\pi^-$ modes of the η cannot be told apart for a pion; together they give the factor 1.047 (v1: 1.056).

Mode	card [%]	HGeant	measured	PDG	factor (v2)	(v1)
$\pi^0 \to \gamma\gamma$	98.798	0.98802	0.98798 ± 0.00152	0.98823	1.000	1.000
$\pi^0 \to e^+e^-\gamma$	1.198	0.011980	0.01202 ± 0.00017	0.01174	0.980	0.977
$\eta \to \gamma\gamma$	37.30	0.37555	0.3769 ± 0.0040	0.3936	1.048	1.044
$\eta \to 3\pi^0$	30.20	0.30406	0.3014 ± 0.0036	0.3257	1.071	1.081
$\eta \to \pi^+\pi^-\pi^0$	21.00	0.21144	0.2074 ± 0.0030	0.2292	1.084	1.105
$\eta \to \pi^+\pi^-\gamma$	4.75	0.047825	0.0497 ± 0.0015	0.0422	0.882	0.849
$\eta \to e^+e^-\gamma$	**6.0**	**0.060410**	0.0640 ± 0.0016	**0.0069**	**0.114**	0.108
$\eta \to \gamma\gamma\pi^0$	0.071	0.000715	–	–	–	–
$\eta \to \pi^+\pi^-e^+e^-$	–	absent	–	0.000268	from the η set	

The weight rules were checked by closure between sets: the same physics taken from two sets must give the same weighted yield. For π^0 Dalitz pairs the lepton/standard ratio is 0.891 ± 0.003 (an unexplained offset of 0.954 times 0.934 from the different Dalitz kinematics of PLUTO and HGeant); for η Dalitz pairs it is 0.879 ± 0.005. Both Dalitz sources thus show a similar deficit; the two values differ by 1.9 standard deviations. With the measured HGeant branching ratio of rules v1 the η value was 0.931, 4 % above the π^0 value. The rows of the η set without a forced-η track agree with the corresponding rows of the standard set to 1.0056 ± 0.0036.

2.5 Cross sections of the components

What the cross sections in this report are. Every σ in Tables 6, 11 and 12 is a *visible* cross section, $\sigma_{\mathrm{vis}} = \sum W/L_{\mathrm{sim}}$: the cross section for producing a *row* (a combination of four tracks) that is reconstructed, passes the PID cuts and the base selection with $z > -160\,\mathrm{mm}$. It is the product of three factors of different origin:

1. the **production cross section** of the mother particle in p+p at 4.5 GeV, taken from the **SMASH model** (Table 9); it is not a measurement and has not yet been compared with one;

2. the **branching ratios** of the decays, **PDG** values (RPP 2020–2024 as quoted in the producers' note), imposed through the weights (Sect. 2.4, Tables 7 and 8); for X17 a hypothesis;

3. the **acceptance times efficiency** (rows per decay), from the **HADES simulation**: HGeant, DST reconstruction, PAT (PID cuts, ranking) and the FAT base selection.

Rows are counted, not events. The expected number of rows in data is $\sigma_{\text{vis}} L_{\text{data}}$, before the data-side corrections (trigger, efficiencies). Table 12 shows the three factors separately for the signal classes; the templates of Table 11 sum over many sources and are given only as totals.

Table 9: Meson production in SMASH (p+p, $T = 4.5\,\text{GeV}$): mean number per collision $\langle n \rangle$ and $\sigma_{\text{prod}} = \langle n \rangle\, \pi b_{\text{max}}^2 = \langle n \rangle \times 45.239\,\text{mb}$. Counted in 200 of the 100 000 converted input files (2×10^7 events, every 500th file), the same files in all sets: η and π^0 directly in the standard set (both are decayed later by HGeant); ρ^0, ω, ϕ (decayed by SMASH) as the forced dilepton decays of the lepton set, which replaces every such meson. Statistical errors. These are model values, not measurements.

Particle	counted in	$\langle n \rangle$ per collision	σ_{prod} [mb]
π^0, all produced by SMASH (without the π^0 of η decays)	std	0.42584	19.265 ± 0.007
of them not from ω or ϕ decays	lep	0.41694	18.862 ± 0.007
η	std	0.023574	1.0665 ± 0.0016
ρ^0	lep	0.0096985	0.4388 ± 0.0010
ω	lep	0.0089169	0.4034 ± 0.0010
ϕ	lep	0.00022265	0.01007 ± 0.00015

Cross-checks of the counts: the 471 482 η of the standard set equal the forced η decays of the η set in the same files (235 656 $\eta \to \pi^+\pi^- e^+ e^-$ + 235 826 $\eta \to \pi^+\pi^- X17$, a split of 49.98 %/50.02 %); the lepton set has 55 forced η fewer, the η of $\phi \to \eta\gamma$ (the ϕ is forced there). The π^0 of the standard set equal those of the lepton set plus the π^0 of ω (177 357) and ϕ (622) decays, within 3.

Truth classes and templates. Every simulated row (a $\pi^+\pi^- e^+ e^-$ combination) is classified from the GEANT truth of its four tracks and the generator codes (`src/sim_truth.h`): the lepton pair gets a class L0–L11, the pion pair a class P0–P9, the four particles together a class 0–8, and the row a template T1–T4 (Table 10). The templates are the coarse split used everywhere: the simulated spectrum is T1+T2, while T3 and T4 are shown only separately. Section 5 splits T2 further into groups G (Table 20).

Table 10: Truth classes of a simulated row (columns `sim_lcls`, `sim_pcls`, `sim_cls`, `sim_tmpl` of `scan_nt`). "Same" means one decay of one mother particle; "FD" a product of a decay forced by the converter. Ghost: a reconstructed track flagged `sim_isghost`.

Lepton pair (`sim_lcls`)	
L0	a track in a lepton slot is not the expected lepton (misidentified)
L1 / L2	both leptons from the same π^0/ η Dalitz decay
L3	both leptons from the same photon conversion
L4 / L5 / L6	$\rho^0, \omega, \phi \to e^+e^-$ / $\omega \to \pi^0 e^+e^-$ / Dalitz decay of the π^0 of $\omega \to \pi^0 e^+e^-$
L7 / L8 / L9	leptons from two different Dalitz decays / a Dalitz decay and a conversion / other different sources
L10 / L11	(η set) leptons of a forced $\eta \to \pi^+\pi^- e^+e^-$ / of the X17
Pion pair (`sim_pcls`)	
P0 / P1	a track in a pion slot is not a pion / a pion that decayed into a muon
P2 / P3 / P4 / P5 / P6	both pions from the same η / ω / ρ^0 / ϕ / K^0_S
P7	both pions primary (SMASH final state, different or no resonance)
P8 / P9	mixed or secondary / (η set) the pions of a forced $\eta \to \pi^+\pi^- e^+e^-$
Four particles (`sim_cls`)	
0 / 1	a misidentified particle / a pion decayed into a muon
2 / 3	bump: $\eta \to \pi^+\pi^-\pi^0$, $\pi^0 \to e^+e^-\gamma$ / the same with a converted π^0 photon
4 / 5 / 6	peak: $\eta \to \pi^+\pi^-\gamma$ with conversion / $\eta \to \pi^+\pi^- e^+e^-$ / $\eta \to \pi^+\pi^- X17$
7 / 8	pions with an e^+e^- pair from one source / leptons from different sources
Template (`sim_tmpl`)	
T1 signal	classes 2–6 (also with a ghost lepton track)
T2 physical background	a true e^+e^- pair from one source (L1–L6; in the η set also L10, L11 outside the signal) with any pions, no fake lepton
T3 fake lepton	L0, a ghost track in a lepton slot, or the same particle in both lepton slots
T4 different sources	L7–L9: neither signal nor background of one source (in data part of the combinatorial background)

Table 11: Templates of the owned $\pi^+\pi^- e^+e^-$ rows ($z > -160\,\text{mm}$), visible cross sections (see the text above). Signal: the four tracks come exactly from one of the signal channels (Table 12). Physical background: a true e^+e^- pair from one source with any pions. Fake lepton: a misidentified track or a ghost track in a lepton slot, or the same particle in both lepton slots. Different sources: the two leptons come from different decays.

	std		lep		eta (pre-rw)	
Template	rows	σ [nb]	rows	σ [nb]	rows	σ [nb]
T1 signal	3 143	14.97	0	–	1 358 905	1.385
T2 physical background	86 834	380.00	6 154 438	303.89	1 622 996	1.518
T3 fake lepton	22 370	99.81	536 261	24.95	165 069	0.351
T4 different sources	3 067	11.96	391 640	0.519	33 428	0.036

Table 12: Signal classes ($z > -160$ mm, all $M(\pi^+\pi^-e^+e^-)$): from production to observed rows, $\sigma_{\rm vis} = \sigma_{\rm prod}(\eta) \times {\rm BR} \times \varepsilon$. $\sigma_{\rm prod}(\eta) = 1.0665$ mb (SMASH, Table 9). BR: PDG; X17 a hypothesis, 10^{-5} for illustration. $N_{\rm gen}$: decays generated in the 10^{10} events; classes 2–4 in the standard set, $N_\eta \times {\rm BR}_{\rm HGeant}$ with the HGeant values of Table 8; classes 5, 6: the forced decays of the η set (counted, Table 9). ε = rows/$N_{\rm gen}$: acceptance times efficiency, for classes 3 and 4 including the probability that the photon converts; it does not depend on any weight. $\sigma_{\rm vis}$ (rules v2): with the decay kinematics of the generators (HGeant phase space, PLUTO for class 5) and with the physical decay kinematics (physics weights, Sect. 3.13; ε changes by 0.889, 0.892, 1.014, 1.023). Errors: statistical, from the number of rows (and 0.15 % from $\sigma_{\rm prod}$). Class 5: the weights assume that exactly half of the η are forced into $\pi^+\pi^-e^+e^-$; the count gives 49.98 %, so the product of the factors is 1.386 instead of the 1.385 nb of the weighted rows. Rules v1 gave 8.05, 5.93, 1.18 and 1.385 nb in the generator column.

			$\sigma_{\rm prod} \times$ BR				$\sigma_{\rm vis}$ [nb]	
Class	Process	BR (PDG)	[nb]	$N_{\rm gen}$	rows	ε	generator	physical
2	$\eta \to \pi^+\pi^-\pi^0$, $\pi^0 \to e^+e^-\gamma$	$0.2292\times$ 0.01174	2 870	5.97×10^5	1 649	2.76×10^{-3}	7.92 ± 0.20	7.05
3	$\eta \to \pi^+\pi^-\pi^0$, $\pi^0 \to \gamma\gamma$, one $\gamma \to e^+e^-$	$0.2292\times$ 0.98823	241 560	4.92×10^7	1 187	2.41×10^{-5}	5.82 ± 0.17	5.19
4	$\eta \to \pi^+\pi^-\gamma$, $\gamma \to e^+e^-$	0.0422	45 005	1.13×10^7	307	2.72×10^{-5}	1.23 ± 0.07	1.24
5	$\eta \to \pi^+\pi^-e^+e^-$	2.68×10^{-4}	285.8	1.18×10^8	571 246	4.85×10^{-3}	1.385 ± 0.003	1.417
6	$\eta \to \pi^+\pi^- X17$, $X17 \to e^+e^-$	10^{-5} (hyp.)	10.66	1.18×10^8	787 659	6.68×10^{-3}	0.0712	0.0712

Three points matter for the use of these numbers:

- The standard set gives the bump and the conversion peak with only 1 649, 1 187 and 307 rows: their statistical errors (2.5–6 %) will limit template fits.
- Conversions are a large part of the simulated bump (42 %) and almost half of the simulated peak.
- The physical background summed over the sets is 685 nb, about forty times the signal (16.4 nb, pre-rw). Most of it is multi-pion production with a π^0 Dalitz or conversion pair, which SMASH does not describe well above the η mass.

2.6 Luminosity for the normalisation to data

The simulation is normalised to the data with $L_{\rm data}$. The value 6080.5 nb^{-1} comes from the producers' note and has not been confirmed by a luminosity analysis of this data set; in this report it is a parameter, and simulated results are also given per nb^{-1} and as shapes. A comparison in absolute units needs, in addition, the data-side corrections (trigger, efficiencies) that the analysis applies anyway.

3 Technical notes for analysers

3.1 Generator codes stored as 32-bit floats

HYDRA keeps the generator codes of every GEANT track exactly, as `Int_t` (`HGeantKine`, `HVirtualCandSim`). PAT, however, stores every quantity of a particle as a `float`, in a map `std::map<std::string,float>` (`core/hcommondef.h`, `core/hparticlecandidate.cc`), and the ntuples are `TNtuples` of `Float_t`. A 32-bit float has a 24-bit significand: integers are exact only up to $2^{24} = 16\,777\,216$. Above that the representable values are spaced by 2 (up to 2^{25}), 4 (up to 2^{26}),

…, 128 (from 2^{30} to 2^{31}), and a code is replaced by the nearest representable value (ties to even). PAT does not warn about it.

The affected columns are `<x>_sim_geninfo` for every particle slot `<x>` (`pip`, `pim`, `ep`, `em`; `ep1`, `ep2`, `em1`, `em2` in the like-sign trees). The columns `<x>_sim_geninfo1` and `<x>_sim_geninfo2` hold small numbers and are exact. Table 13 lists the codes that change.

Table 13: Generator codes above 2^{24} found in these samples and their stored values.

Decay	Set	true code	spacing	stored in the ntuple
$\omega \to \pi^+\pi^-\pi^0$ (SMASH decay)	std, eta	52 080 907	4	52 080 908
$\phi \to \pi^+\pi^-\pi^0$ (SMASH decay)	std, eta	55 080 907	4	55 080 908
$\eta \to \pi^+\pi^- X17$ (forced)	eta	17 700 809	2	17 700 808
$\eta \to \pi^+\pi^- e^+e^-$ (forced)	eta	1 702 030 809	128	1 702 030 848

How the values were identified. (1) All distinct values of the three code columns were listed for each set (in the η set 29, 8 and 17 values). (2) Each value was compared with the dictionary of codes of the converter databases and of the producers' note. (3) A value not found in the dictionary was mapped back by taking all integers n for which `(float)n` equals the stored value and keeping those present in the dictionary. Every value had exactly one candidate: no unknown and no ambiguous value occurred. (4) As a check, the class and weight of all 3 433 534 rows of the η set were recomputed from the stored truth columns, with 0 mismatches. In FAT the mapping is done by `CodeCandidates` in `src/sim_truth.h`; bit 3 of `sim_flags` marks a value outside the dictionary and bit 4 a value with several candidates.

The mapping is not guaranteed to be unique in general. For example, a permuted code 1 702 030 908 would also be stored as 1 702 030 848. Every new dictionary must therefore be checked for such collisions.

Pitfall: the result of a comparison depends on the tool. With a float column `x` holding the stored value 1 702 030 848:

Tool	Expression	rows selected
`TTree::Draw`, `GetEntries`, `Scan` (compares in double)	`x==1702030809`	0
same	`x==1702030848`	1
compiled C++, RDataFrame filter (literal converted to float)	`x==1702030809`	1
C++, comparison with a `double d = 1702030809`	`x==d`	0

Safe practice: compare with the stored value written as a float constant, for example `const float kEtaPPee = 1702030848.f;`, or convert the code explicitly, `x == (float)1702030809`.

Other float32 limits in the chain. The same 24-bit limit applies to every integer that passes through a `Float_t` column. FAT writes the PAT row number (`pat_entry`) as a float, so in simulation it refuses an input tree with more than 2^{24} rows (such a file must be split). PAT stores the run number minus an offset set per beam time and aborts the job if a run is outside $[\text{offset}, \text{offset} + 2^{24})$.

3.2 Which decay a track comes from

The producers' note says that two tracks come from the same forced decay when `geninfo`, `geninfo1` and `geninfo2` are all equal. This is wrong for one important case. `geninfo1` is the code of the direct parent (7051 for an $e^\pm$ of a π^0 Dalitz decay, 7 for its photon), so a Dalitz lepton and a lepton from the conversion of the photon of the *same* decay differ in `geninfo1`. With the triple rule such a pair counts as two decays and gets w^2 instead of w (83 times too small for π^0). The key used in FAT is (`geninfo`, `geninfo2`): the decay code and the counter of the decay in the event.

3.3 The HGeant decay table of the η

In the standard set (and for the non-forced particles of the other sets) the π^0 and the η are decayed by HGeant. The producers measured on 10^6 events that HGeant decays the η into $e^+e^-\gamma$ in 6.4 % of the cases, 9.3 times the PDG value. The cause is in the HGeant configuration of this production:

- Every HGeant input file of the feb22 p+p 4.5 GeV SMASH productions (standard, lepton and η sets; 3200 files, e.g. `/lustre/hades/user/hadesdst/sub/feb22/p4500p/hgeant/p4500p/smash_standard/hgeant/p4500p.dat`, used by the job arrays `gen4_p4500p_smash_*_jobarray.dat` with hgeant2-7.5) contains the card

```
ETA  37.30 30.20 21.00 4.75 6.0 7.1e-2
```

- In the HGeant source (hgeant2-7.4: `geant/hgeakey.F`, `ffkey('ETA',breta,6,'real') ! new eta branchings`; `geant/uginit.F`, decay modes `dmode1` = $\gamma\gamma$, $3\pi^0$, $\pi^+\pi^-\,\pi^0$, $\pi^+\pi^-\,\gamma$, $e^+e^-\gamma$, $\gamma\gamma\pi^0$) the six numbers are the η branching ratios in percent in this order; HGeant normalises them to their sum, 99.321.
- The HGeant default is `39.30 32.20 23.00 4.75 0.60 0.071`. The card raises $\eta \to e^+e^-\gamma$ tenfold (6.0 instead of 0.60 %) and lowers the first three modes by 2.00 each, so that the sum stays close to 100 %. Whether this enhancement was intended must be confirmed by the producers.
- There is no `PI0` card: the π^0 uses the HGeant default, 98.798 and 1.198 %.

The normalised card values agree with the producers' measurement (Table 8). Since rules version 2 the weights use these exact values instead of the measurement; the correction factors change by -1.9 % for $\eta \to \pi^+\pi^-\pi^0$ (the bump), $+3.9$ % for $\eta \to \pi^+\pi^-\gamma$ (the conversion peak), $+5.9$ % for $\eta \to e^+e^-\gamma$ and $+0.3$ % for the π^0 Dalitz decay.

Source of this information: found on 13 November 2026 by reading the HGeant job arrays and input files of the production and the HGeant 7.4 source (SVN checkout `/lustre/hades/user/przygoda/SVN/hgeant2-7.4`). The production ran hgeant2-7.5, whose source was not found; the order of the modes is taken from 7.4 and is confirmed by the agreement with the measured fractions.

3.4 Random rotation of the events in HGeant

The three sets contain the same SMASH events, but a direct comparison of the generator momenta finds almost no common rows (1 of 122 883 pion pairs). The reason is that HGeant rotates every event by a random azimuthal angle around the beam axis, independently in each set: the input files have `SWIT 0 0 0 0 0 1 1 1`, whose sixth switch is the random event plane (comment in the same file: "standard urqmd evtplane+gcalor+neutrocut"). The check:

- The converted files of the three sets (files 1, 13 887, 50 000, 99 999) are identical event by event for every SMASH particle that the converter did not change.
- Matching the rows of the sets by rotation-invariant quantities (p_T and p_z of both generator pions), 5 240 pion pairs of the standard set are found again in the lepton set and 467 in the η set (4.3 % and 0.4 % of its pairs; the rows must pass the selection in both sets). Every match has the same event number, both pions of a match are rotated by the same angle (difference below 10^{-6} rad), and the angles are uniform.

Rows of different sets can therefore be matched only with rotation-invariant quantities and the event number. The statistical errors of comparisons between sets were computed as for independent samples; at the generator level the sets are correlated.

3.5 Several key cycles of one tree

The merged files contain more than one key cycle of the main tree: `ppee_lep_all.root` has `PipPimEpEm_ID;3` (7 622 819 rows) and `;2` (55 454 rows, the first rows of `;3`); `ppee_eta_all.root` has `;2` (3 508 574) and `;1` (33 633). `TChain` and `TFile::Get` without a cycle take the highest cycle, which is the complete tree. Code that loops over the keys of a file must do the same.

3.6 Trigger bit in simulation

In data, PT3 events have `trigbit == 8192`. In simulation the PT3 rows have `trigbit` = 12288 (PT3 and PT2 bits, $8192 + 4096$), so the data condition rejects every simulated row. FAT tests the bit, `(trigbit & 8192) != 0`, when the configuration key `trigger.bit_test` is set.

3.7 Grouping rows into events

Several rows (combinations) belong to one event. In data the event counter restarts in every sub-file, so (`runnumber`, `event`) is not unique; rows are grouped by adjacency of (`file_id`, `runnumber`, `event`). In simulation `runnumber` = 0, and rows are grouped by adjacency of (`event`, `eVertReco_z`). Event grouping is needed for the statistical error of weighted histograms: the variance in a bin is the sum over events of the squared sum of the weights of their rows.

3.8 Converter defect for the second decay

In the lepton set the π^0 of $\omega \to \pi^0 e^+ e^-$ is decayed again ($\pi^0 \to e^+ e^- \gamma$). Its daughters get $w = 0.01198 \times 0.01198$, because the converter multiplies the daughter's own weight by itself, while the probability of the chain is 0.00118×0.01198. Such rows need the factor 0.0985.

3.9 Combinations in the lepton set

Every π^0 of every event of the lepton set is forced into a Dalitz decay, so an event holds on average 0.42 Dalitz pairs, far more than in reality. The choice of the best combination in an event (`isBest`) is therefore not realistic in this set; only the weighted rows describe the decays.

3.10 Precision of weighted sums

Weighted sums over millions of rows must be accumulated in double precision. A `TH1F` filled with the 7 million weighted rows of the lepton set lost 3.4 % of the sum (75 303 instead of 72 804). Use `TH1D` and `double` accumulators.

3.11 Time-of-flight reference in simulation and data processing

The PAT branch of the simulation contains a change (commit `3b8f529`, two-segment calculation of the expected time of flight `tof_mom`) that the data production (tag `prod-feb22-betanew-2026-10-28`) does not have. With the PID polygons refitted on the same footing, the net effect on the selected rows is −0.04 %, so the data were not reprocessed.

3.12 Beam energy in the kinematics

The analysis computes missing masses and the frames with the beam kinetic energy of the configuration, $T = 4530$ MeV, in data and, at present, also in simulation. The SMASH events were generated

at $T = 4500\,\mathrm{MeV}$, which corresponds to a $\sqrt{s}$ lower by about 8 MeV. Quantities that use the beam (missing masses, the centre-of-mass rapidity) are therefore slightly shifted in the simulation. **Open point:** analyse the simulation with $T = 4500\,\mathrm{MeV}$ or quantify the effect on the cuts that use these quantities.

3.13 The $\eta \to \pi^+\pi^- e^+ e^-$ decay of the η set

The $M(e^+e^-)$ distribution of the simulated decay $\eta \to \pi^+\pi^- e^+ e^-$ is not the QED (Kroll–Wada) one. At the generator level, 27 % of the decays have $M(e^+e^-)$ between 1.02 and $1.52\,\mathrm{MeV}/c^2$ and 4.7 % lie above $20\,\mathrm{MeV}/c^2$, against 28 % above $20\,\mathrm{MeV}/c^2$ in QED; the density falls roughly as $1/M^2$. The cause is the converter: it adds the decay as a plain four-body channel (`eta:e+,e-,pi+,pi-` in `database_decays_eta.dat`, via `makeStaticData()->AddDecay`) without a dilepton, i.e. without the virtual photon (compare the Dalitz decays of the lepton set, written as `dilepton[e+,e-]`), so no dilepton model is applied. The PLUTO plugin `eta_physics` is switched off in the converter on purpose: it would impose distributions taken from the DISTO experiment, which are not wanted. The remedy is therefore a channel defined with a dilepton, `eta:dilepton[e+,e-],pi+,pi-`, not the plugin. The consequence for the analysis is a simulated peak with too many pairs at small opening angle: the cut on the e^+e^- opening angle, OA $> 2°$, keeps 0.59 of the simulated peak but 0.78 of the data peak. The X17 decay is correct (a line at $17.000\,\mathrm{MeV}/c^2$).

Recommended shape and simulated shape. The decay proceeds through $\eta \to \pi^+\pi^-\gamma^*$, $\gamma^* \to e^+e^-$. With $q = M(e^+e^-)$ and $M = M(\pi^+\pi^-)$ the physical density is

$$\frac{d\Gamma}{dq\,dM} \propto \underbrace{\frac{2\alpha}{3\pi q}\left(1+\frac{2m_e^2}{q^2}\right)\sqrt{1-\frac{4m_e^2}{q^2}}}_{\text{Kroll–Wada}} \times p^{*3}(M)\,\lambda^{3/2}(m_\eta^2, M^2, q^2)\,|F_V(M^2)|^2\,|F_{\mathrm{VMD}}(q^2)|^2, \quad (1)$$

i.e. about $1/q$ from the threshold $2m_e$, cut off at large q by the kinematics; p^* is the pion momentum in the $\pi^+\pi^-$ rest frame (P wave) and F_V the pion form factor (ρ), which pushes $M(\pi^+\pi^-)$ upwards. The angular distributions are $1+\cos^2\theta_e + (4m_e^2/q^2)\sin^2\theta_e$ for the e^+ in the γ^* rest frame and $\sin^2\theta_\pi$ for the π^+ in the $\pi^+\pi^-$ rest frame.

	$M(e^+e^-) < 1.52\,\mathrm{MeV}/c^2$	$M(e^+e^-) > 20\,\mathrm{MeV}/c^2$	e^+ angle
physics model, Eq. (1), variant V3	about 7 %	about 31 %	$1+\cos^2\theta_e + \ldots$
Kroll–Wada with phase space only		28 %	
simulation (generator, 1 181 801 decays)	27 %[a]	4.7 %	flat

[a] between the threshold 1.02 and $1.52\,\mathrm{MeV}/c^2$.

In the generator the density falls roughly as $1/q^2$, much faster than $1/q$; its pion angle already follows $\sin^2\theta_\pi$. Before a new production the generator should be checked on about 10^5 decays: the share below $1.52\,\mathrm{MeV}/c^2$ should be about 7 %, above $20\,\mathrm{MeV}/c^2$ about 30 %, and the e^+ angle must not be flat.

Physics weights. No new production is needed: each simulated row gets a weight that replaces the generator's decay distribution by a physics model (`plots/eta_sim/sim_eta_physics.h`).

- Direct decay: $w = T/G$, with G the generator density fitted to 1 181 801 generated decays and T = Kroll–Wada × the P-wave $\pi^+\pi^-$ γ^* rate $p^{*3}\lambda^{3/2}$ × $|F_V|^2$ (pion vector form factor) × $|F_{\mathrm{VMD}}|^2$ × the decay angles of the e^+ and the π^+ (main variant V3). The mean weight over the generated decays is 1, so the branching ratio is unchanged. The model gives $\Gamma(\pi^+\pi^-e^+e^-)/\Gamma(\pi^+\pi^-\gamma)$ within 1 % of the PDG value.
- Bump: the measured Dalitz-plot distribution of $\eta \to \pi^+\pi^-\pi^0$ (KLOE-2), since HGeant decays it with pure phase space (checked in the HGeant source).

- Conversion peak: $|F_V|^2 p^{*2} k^2$ for $\eta \to \pi^+\pi^-\gamma$, also phase space in HGeant.

Effect in the base selection: the share of the peak with $M(e^+e^-) > 20\,\mathrm{MeV}/c^2$ goes from 0.046 to 0.219 (data 0.280 ± 0.071); the OA $> 2^\circ$ efficiency of the peak goes from 0.59 to 0.697 ± 0.015 (data 0.778 ± 0.033): the tension is reduced, not removed. The details, the evidence for the generator problem and the recommendations to the simulation producers are in a separate note in Polish (`docs/SYMULACJA_ETA_GENERATOR.md`).

4 The analysis in brief

4.1 Hypotheses and combinations

PAT builds, in each event, every combination of tracks that fits a hypothesis with the right charges: $\pi^+\pi^- e^+e^-$ (`PipPimEpEm_ID`) for the signal, and the like-sign $\pi^+\pi^-$ e^+e^+ and $\pi^+\pi^-$ e^-e^- for the combinatorial background in data, $\mathrm{CB} = 2\sqrt{N_{++}N_{--}}$ times a correction factor k determined from data. A lepton slot takes only tracks with a correlated RICH ring (`HParticleCand::getRingCorr()`); a pion slot takes the tracks without one. In simulation PAT also drops tracks without a GEANT identity. All particles of a combination must pass the PID cut of their slot (Sect. 4.2). The combinations of an event are ranked by a χ^2 of the times of flight; the best has isBest $= 1$, the others isBest $= 0$. This analysis keeps all combinations, isBest ≥ 0.

In simulation no combinatorial background is subtracted. The signal of a channel is the set of rows whose four tracks come, according to the GEANT truth and the generator codes, exactly from that production and decay channel; this is the equivalent of the data signal after CB subtraction. The like-sign trees of the simulation are used only for side studies (combinatorics, `isBest`).

4.2 Particle identification with β_{new}

The PID uses the plane $(p, \beta_{\mathrm{new}})$: the momentum from tracking and a velocity reconstructed without a start detector, from the relative times of flight of the tracks of the hypothesis. The cuts are polygons around the bands of p, π^+, π^-, e^+ and e^-, fitted at several widths (1σ to 5σ) on PID samples; the productions use 3σ for p and $\pi^\pm$ and 5σ for $e^\pm$. The experiment cuts (`betanew_v1`) were fitted on data, the simulation cuts (`betanew_v6`) on the SMASH simulation with the GEANT truth of all particles of the hypothesis. In both files the two lepton cuts are identical (experiment: the e^- cut is a copy of the e^+ cut; simulation: the e^+ cut is a copy of the e^- cut).

Figure 1 is the summary of the PID: simulated tracks before the PID cut, each measured in a slot whose mass hypothesis is its own kind, with the cuts used in the simulation (solid) and in the experiment (dashed). Protons and pions come from the hadron PID sample (hypothesis $\mathrm{p}\pi^+\,\pi^-$), electrons and positrons from the lepton PID sample (hypothesis $\pi^+\pi^- e^+e^-$); neither sample has a trigger condition. Figure 2 enlarges the lepton region. The two sets of cuts differ little: the simulated bands are slightly narrower.

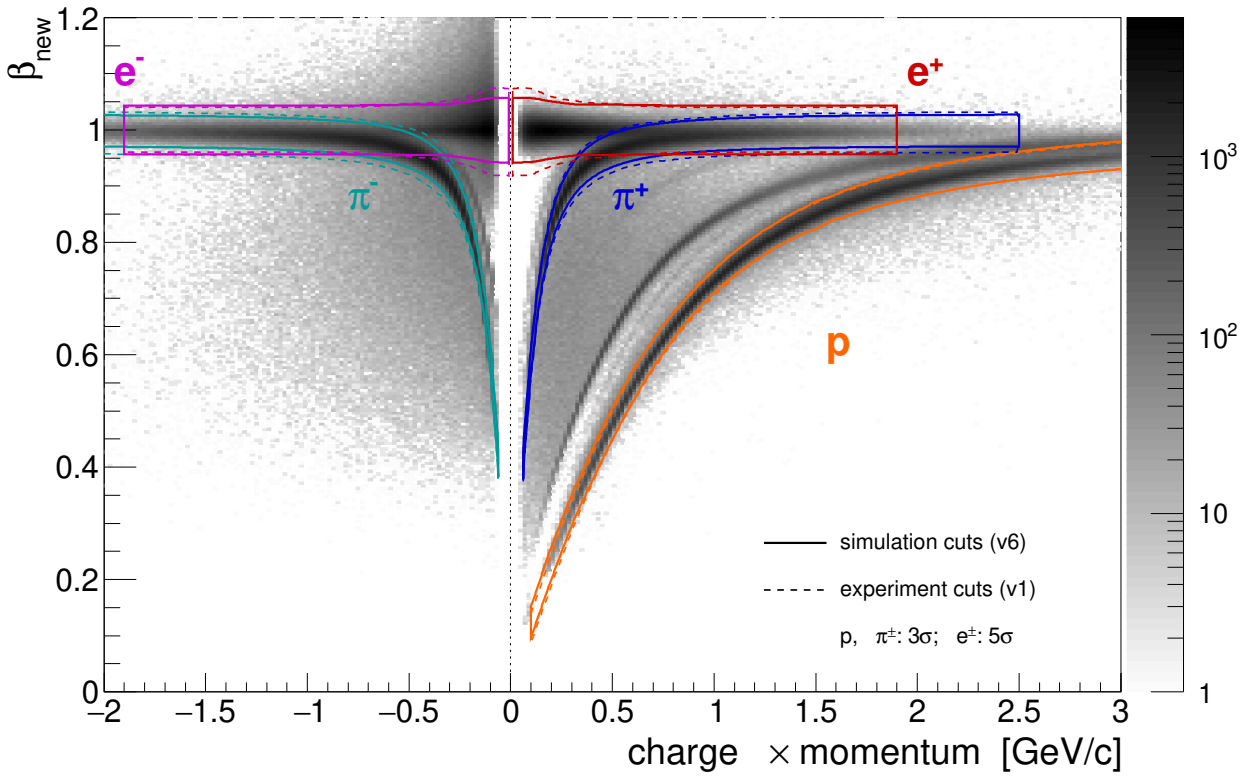


Figure 1: β_{new} versus charge times momentum for simulated tracks before the PID cut (SMASH p+p at 4.5 GeV, no trigger condition, isBest = 1, vertex $z > -500$ mm). Protons and pions: hadron PID sample, slots of the hypothesis $p\pi^+\,\pi^-$ (`HH_elossfix`, 150 of 1000 files); $e^{\pm}$: lepton PID sample, lepton slots of the hypothesis $\pi^+\pi^- e^+ e^-$ (`LEP_merged`, 101 files). Solid lines: simulation cuts (v6), dashed: experiment cuts (v1); p and $\pi^{\pm}$ at 3σ, $e^{\pm}$ at 5σ. The weak band between the π^+ and the proton bands is made of protons in the π^+ slot and of π^+ in the proton slot (see the text).

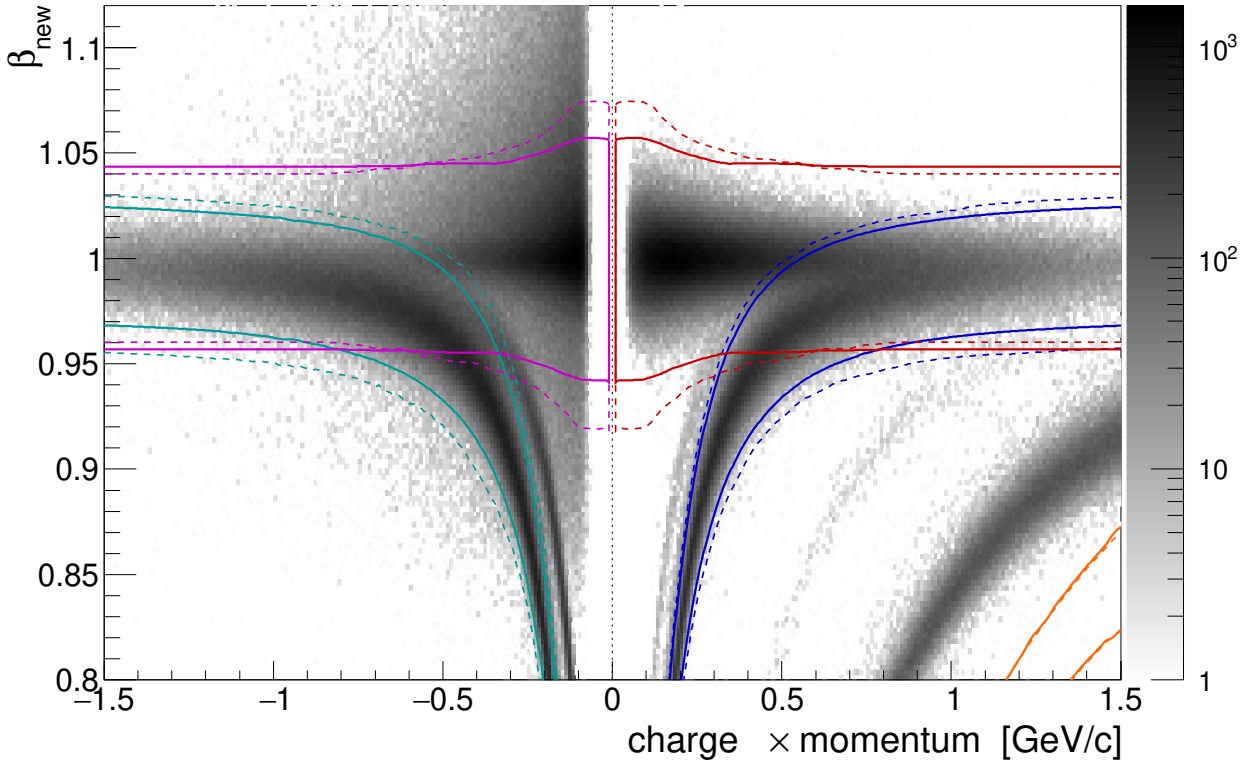


Figure 2: The lepton region of the $\pi^+\pi^-e^+e^-$ hypothesis before the PID cut (all four slots, lepton PID sample). Above about 0.4 GeV/c the pion bands enter the $e^\pm$ polygons: there leptons are separated from pions by the RICH ring, not by β_{new}. Lines as in Fig. 1.

Electrons without a RICH ring in the π^- slot. In Fig. 2 a second band runs to the right of the π^- band, almost parallel to the π^- cut. It is made of electrons in the π^- slot (Fig. 3). A lepton slot takes only tracks with a correlated RICH ring (Sect. 4); an electron without one is treated as a hadron and can occupy the π^- slot. Measured with the pion mass, it follows Eq. (3) derived below, $1/\beta_{\text{new}} = \frac{1}{2}(1/\beta_e + 1/\beta_\pi)$. The most probable β_{new} of these electrons is 0.76, 0.90 and 0.97 at $p = 0.1$, 0.2 and 0.4 GeV/c (expected 0.74, 0.90, 0.97), while the π^- band lies at 0.61, 0.82 and 0.94. Muons in the π^- slot (5 %) are shifted much less (0.846 against 0.821 at 0.2 GeV/c) and lie mostly inside the π^- cut.

These electrons are not a small effect: before the PID cut they are 16 % of the π^- slot (72 769 of 456 971 tracks), and the π^- cut keeps 23 % of them. This agrees with the 4.3 % of the rows of the lepton-set `_ID` tree that have an electron in the π^- slot (bit 5 of `sim_flags`). Table 14 compares them with the electrons that have a ring.

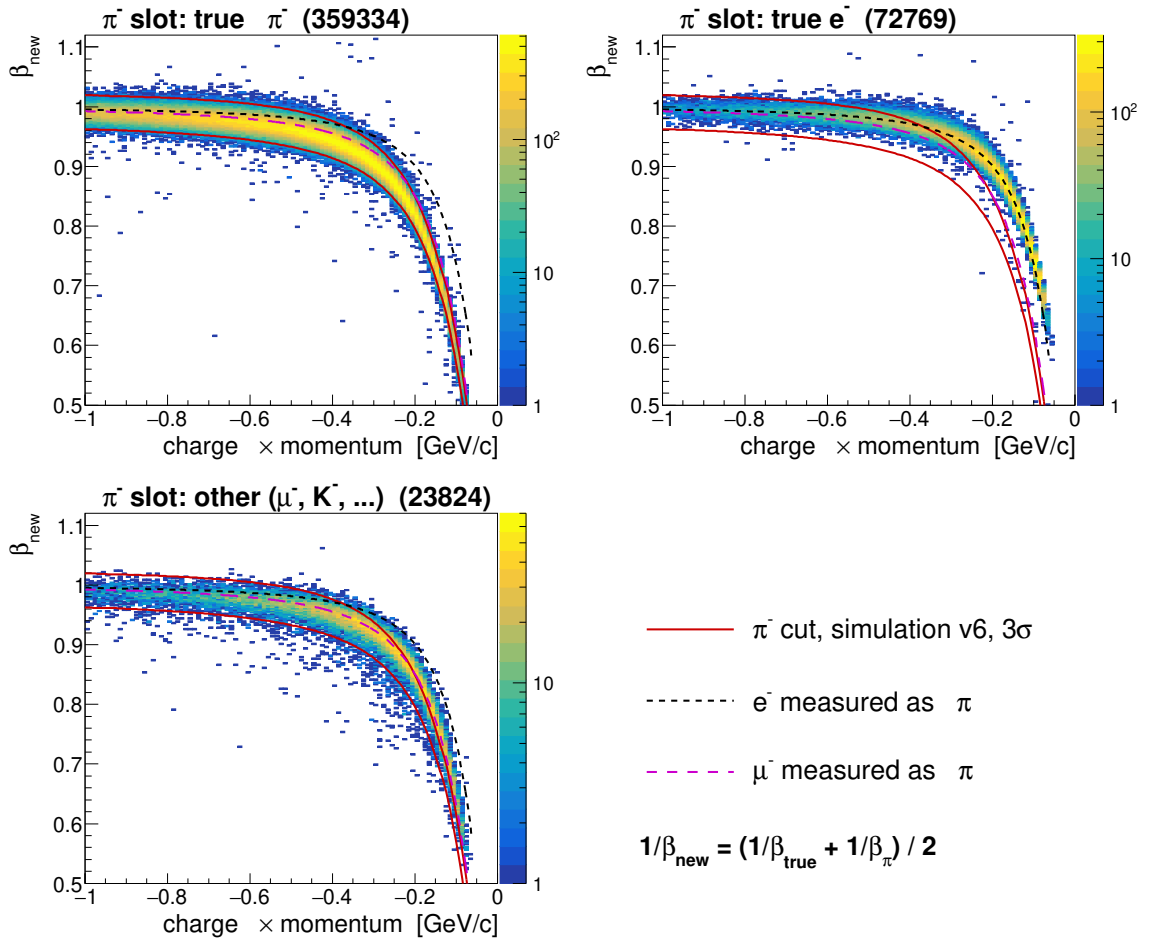


Figure 3: The π^- slot of the $\pi^+\pi^-e^+e^-$ hypothesis before the PID cut (lepton PID sample, isBest = 1, vertex $z > -500$ mm), split by the true particle: π^-, e^-, and the rest (mostly μ^-). Red: π^- cut (simulation v6, 3σ). Dashed: β_{new} expected for an electron (black) and a muon (magenta) measured with the pion mass, Eq. (3).

Table 14: Electrons without a RICH ring (in the π^- slot) and electrons with a ring (in the e^- slot): GEANT creation process, parent particle and creation place, in percent of each group. Lepton PID sample without PID cut, hypothesis $\pi^+\pi^-e^+e^-$, isBest = 1, vertex $z > -500$ mm, 25 of the 101 files (18 011 and 113 135 electrons). Target region: $-200 < z < -60$ mm and $r < 20$ mm. Macros `macros/norich.C`, `macros/norich_zr.C`; numbers in `tables/norich.txt`.

	e^- in the π^- slot (no ring)	e^- in the e^- slot (ring)
photon conversion ($\gamma \to e^+e^-$)	86.6	35.2
Compton scattering	9.3	0.3
knock-on (delta) electron	1.8	0.2
decay (Dalitz decays of π^0, η)	2.3	64.4
parent: photon / π^0/ η	96.0 / 1.6 / 0.7	35.4 / 46.5 / 17.9
created in the target region	4.1	91.5
created at $z = 200$–500 mm	73.9	0.1
created off the beam axis ($r > 20$ mm)	95.8	7.7
ring found (`isring`) / lepton flag (`kIsLepton`)	8.7 / 4.2	100 / 99.6
ghost track (`sim_isghost`)	29.2	8.9
polar angle $\theta < 18°$	19.0	3.8

The electrons without a ring are secondary electrons of photons (mostly from π^0 decays) produced in the detector material *downstream of the RICH radiator*:

- 28 % of their creation points lie within ±6 mm of a sphere of radius 872 mm centred at $z \approx -625$ mm. This is the RICH mirror: in the HGeant geometry (`rich.geo`) its radius is 871–873 mm and its centre is at $z = -532.2 - 88.5 = -620.7$ mm. These are conversions in the mirror itself (carbon and glass segments).
- Most of the others are created at $z \approx 350$–500 mm and $r \approx 150$–350 mm, about 0.6 m from the target, which is the region of the first drift chambers (MDC I). This attribution is inferred from the position; the HGeant medium numbers were not translated into material names.

They do not cross the C_4F_{10} radiator, so they emit no Cherenkov light there: the missing ring is expected and is not an inefficiency of the RICH. They still leave inner MDC segments and are reconstructed as tracks.

Charge asymmetry. Ring-less positrons are rare: 0.6 % of the π^+ slot against 16 % of electrons in the π^- slot. This is not an effect of the hypothesis topology. In the hadron PID sample without PID cut (hypothesis $\pi^+\ \pi^-$, *all* combinations of 542 211 events, every distinct ring-less track counted once per event) the tracks created outside the target region give $e^+ : e^- = 2\,633 : 24\,751$, a ratio of 0.106 (Table 15). The ratio depends little on the momentum but strongly on the polar angle. Of the conversion pairs, the positron partner of a reconstructed electron is found in only 4.4 % of the cases, the electron partner of a reconstructed positron in 38 %; both charges have the same ghost fraction (30–32 %), so the pair is not merged into one track: the positron is lost. This agrees with the acceptance in the toroidal field: positive particles are bent towards small polar angles and leave the acceptance, negative ones are bent outwards. The strong dependence on θ supports this; the asymmetry that remains at large θ and high momentum is not yet fully understood.

Table 15: Ring-less e^+ (in the π^+ slot) and e^- (in the π^- slot) created outside the target region, hadron PID sample without PID cut (`HH_elossfix`, 60 files, all combinations, each track counted once per event; macro `macros/asym.C`).

true p [MeV/c]	e^+	e^-	e^+/e^-	θ [deg]	e^+	e^-	e^+/e^-
0–100	302	1 843	0.164	< 20	157	6 814	0.023
100–200	750	8 403	0.089	20–30	585	6 654	0.088
200–300	564	5 086	0.111	30–50	1 125	7 974	0.141
300–500	551	5 202	0.106	50–90	766	3 309	0.231
> 500	466	4 217	0.111				
all	2 633	24 751	0.106				

Which cuts remove them. After the PID cut, in the analysis ntuple of the standard set (rules v2, base selection, $z > -160$ mm), electrons are 2.5 % of the π^- slot (5 464 rows); almost all have $p > 300$ MeV/c, where the β_{new} bands of electrons and pions overlap, 95 % are created at $z > 100$ mm and 90 % are conversions. Table 16 gives the fraction of pions and electrons kept by candidate cuts.

- **Vertex cuts do not help:** a conversion electron continues in the direction of its photon, which came from the target, so its track points back to the vertex. The use in the vertex fit, the distance $|z_{\text{trk}} - z_{\text{vtx}}|$ and the distance r from the beam axis keep electrons as often as pions.
- **The energy loss in the MDC separates them:** median 2.75 for these electrons against 1.56 for pions. A cut at 2.2 keeps 91.5 % of the pions and 15 % of the electrons, and lowers the electron share of the π^- slot from 2.5 to 0.4 %. Below 300 MeV/c it keeps only 79 % of the pions (their energy loss rises), so a momentum-dependent cut is preferable.

- Track χ^2 and the angle to the nearest other track remove only about a quarter of them.

Not yet checked: whether the MDC energy loss of simulation agrees with data, and whether the high energy loss of these electrons is the relativistic rise or the double deposit of a close pair.

Table 16: Fraction of true π^- and of electrons in the π^- slot kept by candidate cuts, analysis ntuple `scan_nt` of the standard set (rules v2, after the PID cut, $z > -160$ mm; macro `macros/eminpim_cuts.C`; the lepton set gives the same picture). Numbers in `tables/eminpim.txt`.

	kept: π^- / e^-			
Cut	$p < 300$ MeV/c	300–600 MeV/c	≥ 600 MeV/c	all
rows (π^- / e^-)	54 891 / 67	85 475 / 3 508	74 669 / 1 889	215 035 / 5 464
track used in the vertex fit				0.932 / 0.947
$\lvert z_{\rm trk} - z_{\rm vtx}\rvert < 20$ mm				0.860 / 0.834
$r < 10$ mm				0.938 / 0.949
MDC $dE/dx < 2.2$	0.787 / 0.373	0.965 / 0.165	0.951 / 0.105	0.915 / 0.147
MDC $dE/dx < 2.0$	0.679 / 0.239	0.921 / 0.058	0.891 / 0.035	0.849 / 0.053
track $\chi^2 < 50$	0.886 / 0.851	0.920 / 0.769	0.949 / 0.790	0.921 / 0.777
no track within 2°	0.939 / 0.716	0.949 / 0.743	0.962 / 0.739	0.951 / 0.742

$\beta_{\rm new}$ depends on the mass hypothesis of the track. The relative-time method (`HTimeCut`) takes one track r of the combination as the time reference: an e^- if there is one, otherwise a proton, and among several candidates the one with the smallest χ^2. For every other track j it uses the time

$$t_j^{\rm new} = t_j - \tfrac{1}{2}\Big[(t_r - t_r^{\rm exp}) + (t_j - t_j^{\rm exp})\Big], \qquad \beta_{\rm new} = \frac{L_j}{c\, t_j^{\rm new}}, \tag{2}$$

where t is the measured time of flight, $t^{\rm exp} = L/(c\,\beta(p,m))$ the time expected for the mass m assumed in the hypothesis, and L the path length. The start-time correction is thus the average of the offsets $t - t^{\rm exp}$ of the reference and of the track itself. For a track of the assumed mass the method gives its true velocity. For a track of another mass half of the difference is absorbed; if the reference is correct,

$$\frac{1}{\beta_{\rm new}} = \frac{1}{2}\left(\frac{1}{\beta_{\rm true}} + \frac{1}{\beta_{\rm hyp}}\right). \tag{3}$$

A proton of 1 GeV/c in a π^+ slot has $\beta_{\rm new} = 0.84$ instead of 0.73. A π^+ in a proton slot lands on the same curve, which is the weak middle band of Fig. 1. The reference track itself gets $t_r^{\rm new} = t_r - \frac{1}{2}(t_r - t_r^{\rm exp}) - \frac{1}{2}\langle t_j - t_j^{\rm exp}\rangle_{j\neq r}$: half of its own offset and half of the mean offset of its partners, so a wrongly assigned partner shifts the velocity of the reference.

Figure 4 shows what the PID cut acts on in this analysis: all four slots of the $\pi^+\pi^- e^+ e^-$ hypothesis. The protons, which can only occupy the π^+ and e^+ slots, form the band of Eq. (3) between the pion and the proton bands; only 7 % of them lie inside the proton cut, and 2.6 % inside the π^+ cut. Figure 5 splits these tracks by their true identity, and Table 17 gives the composition of each slot before the PID cut together with the fraction of each kind of track inside the cut of the slot.

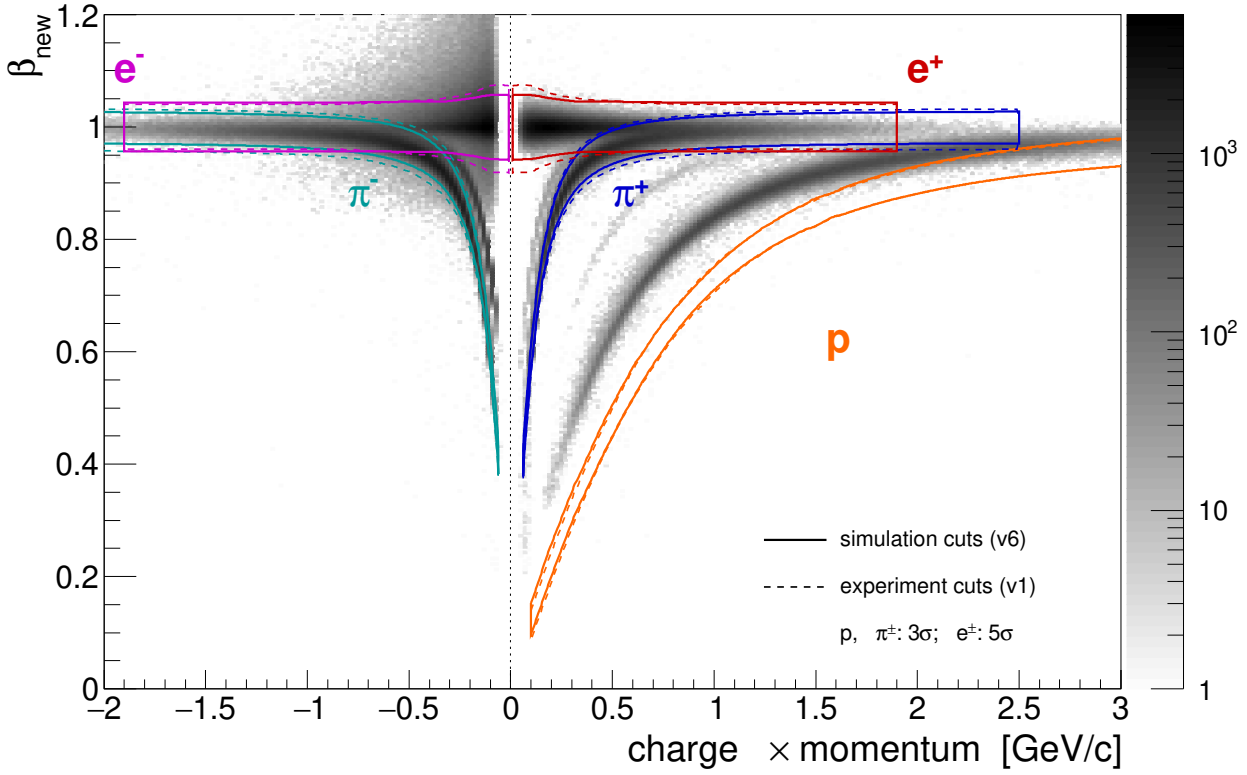


Figure 4: All four slots of the $\pi^+\pi^-e^+e^-$ hypothesis before the PID cut (lepton PID sample, isBest = 1, vertex $z > -500\,\mathrm{mm}$). The protons appear in the π^+ and e^+ slots, measured with the pion or electron mass, halfway between the pion and the proton bands. Lines as in Fig. 1.

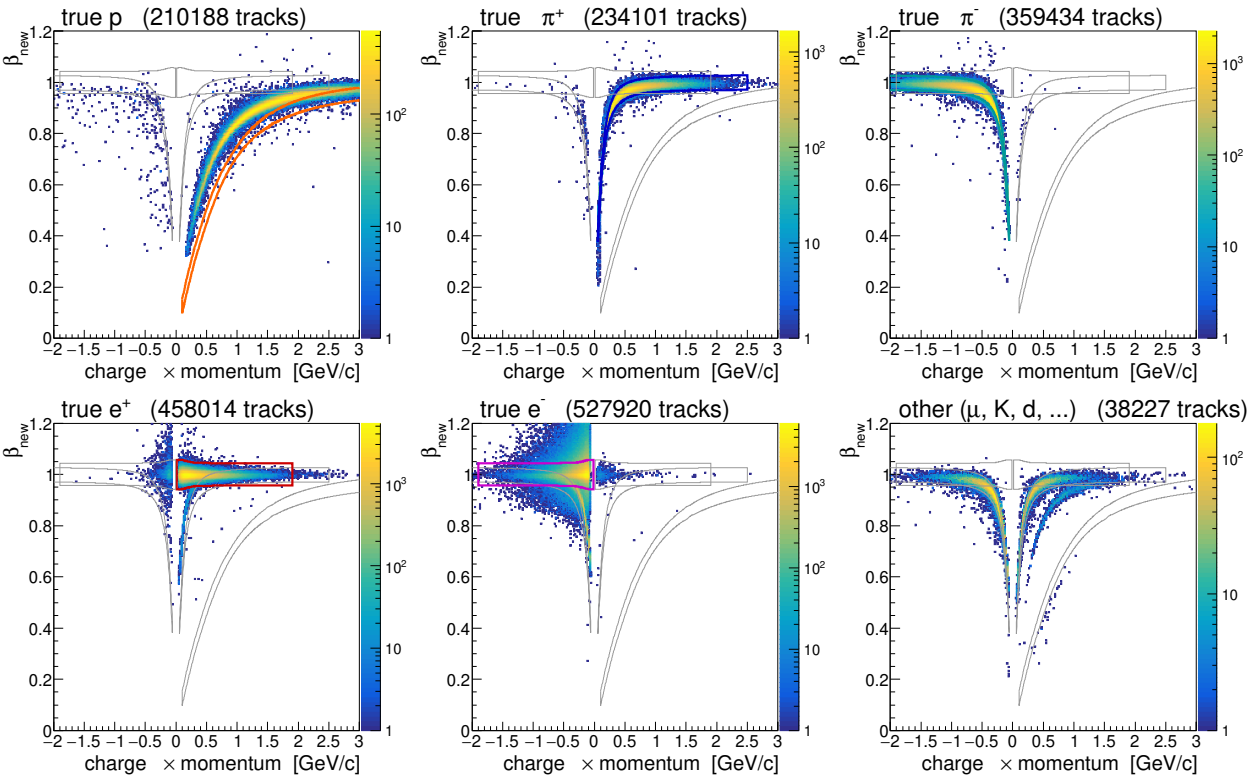


Figure 5: The tracks of Fig. 4 split by their true identity (GEANT). The coloured polygon is the simulation cut of that species, the grey ones the other cuts.

Table 17: Composition of the slots before the PID cut and the fraction of each kind of track that lies inside the simulation cut of its slot (v6; p, $\pi^{\pm}$ 3σ, $e^{\pm}$ 5σ). Each cell: share of the slot's tracks / fraction inside the cut. Simulation PID samples, isBest = 1, vertex $z > -500$ mm, no trigger condition. Shares below 0.1 % are omitted.

Hypothesis: slot	tracks	p	π^+	π^-	e^+	e^-	other
$p\pi^+\pi^-$: p	642804	0.889 / 0.936	0.104 / 0.010	–	–	–	0.007 / 0.019
$p\pi^+\pi^-$: π^+	642804	0.164 / 0.065	0.781 / 0.837	–	0.005 / 0.228	–	0.049 / 0.567
$p\pi^+\pi^-$: π^-	642804	–	–	0.916 / 0.848	–	0.032 / 0.254	0.051 / 0.636
$\pi^+\pi^-e^+e^-$: π^+	456971	0.453 / 0.026	0.510 / 0.988	–	0.006 / 0.345	–	0.031 / 0.682
$\pi^+\pi^-e^+e^-$: e^+	456971	0.005 / 0.013	0.002 / 0.742	–	0.991 / 0.999	0.002 / 0.989	–
$\pi^+\pi^-e^+e^-$: π^-	456971	0.002 / 0.313	–	0.786 / 0.985	–	0.159 / 0.228	0.052 / 0.668
$\pi^+\pi^-e^+e^-$: e^-	456971	–	–	–	0.005 / 0.840	0.994 / 0.837	–

What the table shows:

- In the $\pi^+\pi^-e^+e^-$ hypothesis the reference is the e^-, and the e^- slot is 99.4 % pure. Correctly assigned pions and positrons then lie inside their cuts at the level expected from the cut widths: 98.8 % (π^+), 98.5 % (π^-), 99.9 % (e^+).
- The e^- itself, being the reference, is inside its cut only at 83.7 %: before the PID cut 45 % of the π^+ slots hold a proton, and such a partner shifts the time of the reference. After the PID cut, when all partners must be inside their bands, this effect is much smaller.
- In the $p\pi^+$ π^- hypothesis there is no e^-, so the proton slot is the reference. With 10 % π^+ in the proton slot and 16 % protons in the π^+ slot, even correctly assigned tracks lie inside their cuts at only 84–94 %.
- Wrongly assigned tracks are mostly rejected: protons in a π^+ slot pass the π^+ cut at 2.6 % ($\pi^+\pi^-e^+e^-$) and 6.5 % ($p\pi^+$ π^-), π^+ in a proton slot pass the proton cut at 1.0 %.

The PID cut efficiency of a track therefore depends on its partners. Efficiencies have to be taken from complete combinations, as the analysis does with the `_ID` trees and the GEANT truth, and not from the width of single-track bands.

4.3 Selection

The base selection ("pre") is common to all studies:

1. all combinations that passed the PID cuts, isBest ≥ 0;
2. a reconstructed event vertex, $z > -500$ mm (mandatory);
3. the PT3 trigger (data: `trigbit == 8192`; simulation: bit test);
4. the target region, $z > -160$ mm (the target spans -152 to -98 mm at half maximum, the maximum is at -131 mm);
5. momenta corrected for the energy loss for all four particles (in data the corrected π^- momentum is missing for 0.7 % of the rows, π^- at $\theta < 16°$; such rows are dropped).

On top of it the analysis studies the cuts of Table 18. Each of them is stored in `scan_nt` both as a value and as a pass flag, so that every cut can be scanned and its effect shown separately.

Table 18: Cut variables of the analysis. All are computed from the corrected momenta. The values are those of the named selections used so far; the cut scan of the data is summarised in a later part.

Variable	Meaning	Values used
OA	opening angle of the e^+e^- pair (lab)	$> 2^\circ$ (conversions)
α_{lab}	angle between the $\pi^+\pi^-$ and the e^+e^- systems (lab)	$< 50^\circ$, 55, 40
$M(\pi^+\pi^-)$	invariant mass of the pions	< 0.42, < 0.46, $> 0.30\,\text{GeV}/c^2$
α_η	angle between $\pi^+\pi^-$ and e^+e^- in the rest frame of a hypothetical η with the measured $\pi^+\pi^-e^+e^-$ momentum and the mass m_η (a band-pass in $M(\pi^+\pi^-e^+e^-)$)	$> 140^\circ$
pions in vertex	both pions used in the event-vertex fit (`isUsedVertex`)	flag
Δz_π	$\max(\lvert z_{\pi^+} - z_{\text{vertex}}\rvert, \lvert z_{\pi^-} - z_{\text{vertex}}\rvert)$, z of a track at its closest approach to the beam axis	$< 40\,\text{mm}$
MM_4, $p_{T,4}$	missing mass of $\pi^+\pi^-e^+e^-$, the mass of (beam + target $-$ $\pi^+\pi^-e^+e^-$) four-momenta; transverse momentum of $\pi^+\pi^-e^+e^-$ (lab)	studied

Table 19: Named selections (all include "pre").

Name	Cuts in addition to "pre"
pre+OA> 2	OA $> 2^\circ$
slides	$\alpha_{\text{lab}} < 50^\circ$, $M(\pi^+\pi^-) < 0.46\,\text{GeV}/c^2$, $\alpha_\eta > 140^\circ$, OA$> 2^\circ$
old code	$\alpha_{\text{lab}} < 50^\circ$, $M(\pi^+\pi^-) < 0.42\,\text{GeV}/c^2$, $\alpha_\eta > 140^\circ$
peak nominal	$\alpha_{\text{lab}} < 55^\circ$, $M(\pi^+\pi^-) > 0.30\,\text{GeV}/c^2$
bump nominal	$\alpha_{\text{lab}} < 40^\circ$, $0.30 < M(\pi^+\pi^-) < 0.41\,\text{GeV}/c^2$

4.4 Software chain

1. **PAT**, PostDST Analysis Tool (`git.gsi.de/przygoda/pat3`): DST → hypothesis ntuples with PID, relative-time reconstruction and ranking; simulation branch `pp45_sim_pippimepem`.
2. **FAT**, Final Analysis Tool (this repository, branch `pp45_pippimepem_betanew_sim`): base selection, kinematics, truth classes and weights (`src/sim_truth.h`), ntuple `scan_nt`; configurations `config_sim_{std,lep,eta}*.json`.
3. **Macros** (`plots/eta_sim/`): the common engine `SimEngine.h` (compact files, components, presets, weighted spectra with event-grouped errors) and the studies; figures in `plots/eta_sim/figs/`, tables in `plots/eta_sim/tables/`.

5 Simulation study: from the signal to the selection

This section follows the order planned for the study (signal, acceptance, cuts, backgrounds, optimisation, test of the extraction). It uses **simulation only**: no measured data were read. The simulation contains no combinatorial background (CB), and none was added. All statements about cuts therefore concern the physical background of SMASH, not the CB of the data.

Common settings. Every number of this section uses the same rows and weights:

- rows of the $\pi^+\pi^-e^+e^-$ trees of the three sets (std, lep, η), owned in mode A (Sect. 2.4);
- the weight W of Sect. 2.4 (PDG branching ratios) times the physics weights of the η decays, main variant V3 (Sect. 3.13); for the bump the KLOE-2 Dalitz plot, for the conversion peak $|F_V|^2 p^{*2} k^2$. All numbers of this section use the weight rules v2 (exact HGeant branching ratios, Sect. 3.3);
- the base selection "pre" (Sect. 4.3); all cut variables from the corrected momenta;
- cross section $\sigma = \sum W / L_{\text{sim}}$ with $L_{\text{sim}} = 221.05\,\text{nb}^{-1}$;
- statistical errors from event groups: the variance of a sum is the sum over events of the squared sum of the weights of their rows. The effective number of rows, $N_{\text{eff}} = (\sum W)^2/\text{Var}$, is quoted where it is small;
- where a number of rows in data is needed (significance, expected limits), the luminosity $L_{\text{data}} = 6080.5\,\text{nb}^{-1}$ (UNCONFIRMED) is used as a scale only: one unit of $\sum W$ corresponds to 27.51 expected rows.

The simulated sum is template T1 (signal classes 2–5) plus template T2 (physical background). Templates T3 (fake lepton) and T4 (leptons from two sources) are never added; they are shown separately and enter a background only as a stated variant. X17 rows (class 6) carry $W = 2\,\text{BR}$ per row, BR is a free parameter, and they are never added to the sum. The windows in $M(\pi^+\pi^-e^+e^-)$ and the background groups are the same in all parts (Table 20).

Table 20: Windows in $M(\pi^+\pi^-e^+e^-)$ and background groups used in this section. The peak window is centred on the simulated peak ($0.5460 \pm 1.9 \times 0.0067\,\text{GeV}/c^2$); the bump window is the data bump window scaled by the simulated mass scale 1.0057. The groups split template 2 by the lepton-pair class (L) and the pion-pair class (P) of `src/sim_truth.h` (all classes and the templates T1–T4 are defined in Table 10); a row goes to the first group in the order given whose condition it fulfils.

Name	Definition
peak window	$0.5333 \le M(\pi^+\pi^-e^+e^-) < 0.5587\,\text{GeV}/c^2$
bump window	$0.4576 \le M(\pi^+\pi^-e^+e^-) < 0.5255\,\text{GeV}/c^2$
other regions	below 0.4576; gap 0.5255–0.5333; 0.5587–0.70; 0.70–1.00; above $1.00\,\text{GeV}/c^2$
classes 2, 3, 4, 5	the signal classes (Table 12): bump Dalitz, bump conversion, peak conversion, direct decay
G_η	template 2 with both pions from one η (P2) or an η Dalitz pair (L2): an η is produced, but the row is not a signal combination
$G_{\pi^0\text{D}}$	remaining template-2 rows with a π^0 Dalitz pair (L1)
G_{conv}	remaining template-2 rows with a conversion pair (L3)
G_{vec}	remaining template-2 rows with $\rho^0, \omega, \phi \to e^+e^-$, $\omega \to \pi^0 e^+e^-$ or a second-decay pair (L4–L6)
G_{oth}	all other template-2 rows (in practice the e^+e^- pair of a direct $\eta \to \pi^+\pi^-e^+e^-$ decay with pions from elsewhere)
G_{fake}, G_{diff}	template 3 and template 4 (variants only)
multi-pion	$G_{\pi^0\text{D}} + G_{\text{conv}} + G_{\text{vec}}$

5.1 The full spectrum and the cut sequence

The spectrum after "pre" (Fig. 6). The simulated sum is $\sigma(\text{T1} + \text{T2}) = 701.1 \pm 1.5\,\text{nb}$ over all $M(\pi^+\pi^-e^+e^-)$. Of this, the four signal classes are $14.89 \pm 0.27\,\text{nb}$ and the multi-pion background $659.9 \pm 1.4\,\text{nb}$ (94 %): $G_{\pi^0\text{D}} = 394.5 \pm 0.8\,\text{nb}$, $G_{\text{conv}} = 265.3 \pm 1.2\,\text{nb}$ and $G_{\text{vec}} = 0.107 \pm 0.001\,\text{nb}$. G_η

is 24.27 ± 0.21 nb and G_{oth} 1.967 ± 0.005 nb. Not in the sum: T3 125.0 ± 0.7 nb, T4 12.51 ± 0.25 nb. Above the kinematic limit $M(\pi^+\pi^-e^+e^-) = \sqrt{s} - 2m_p = 1.591\,\text{GeV}/c^2$ there are 2.87 nb (0.4 %); 2.58 nb of them contain a misidentified track.

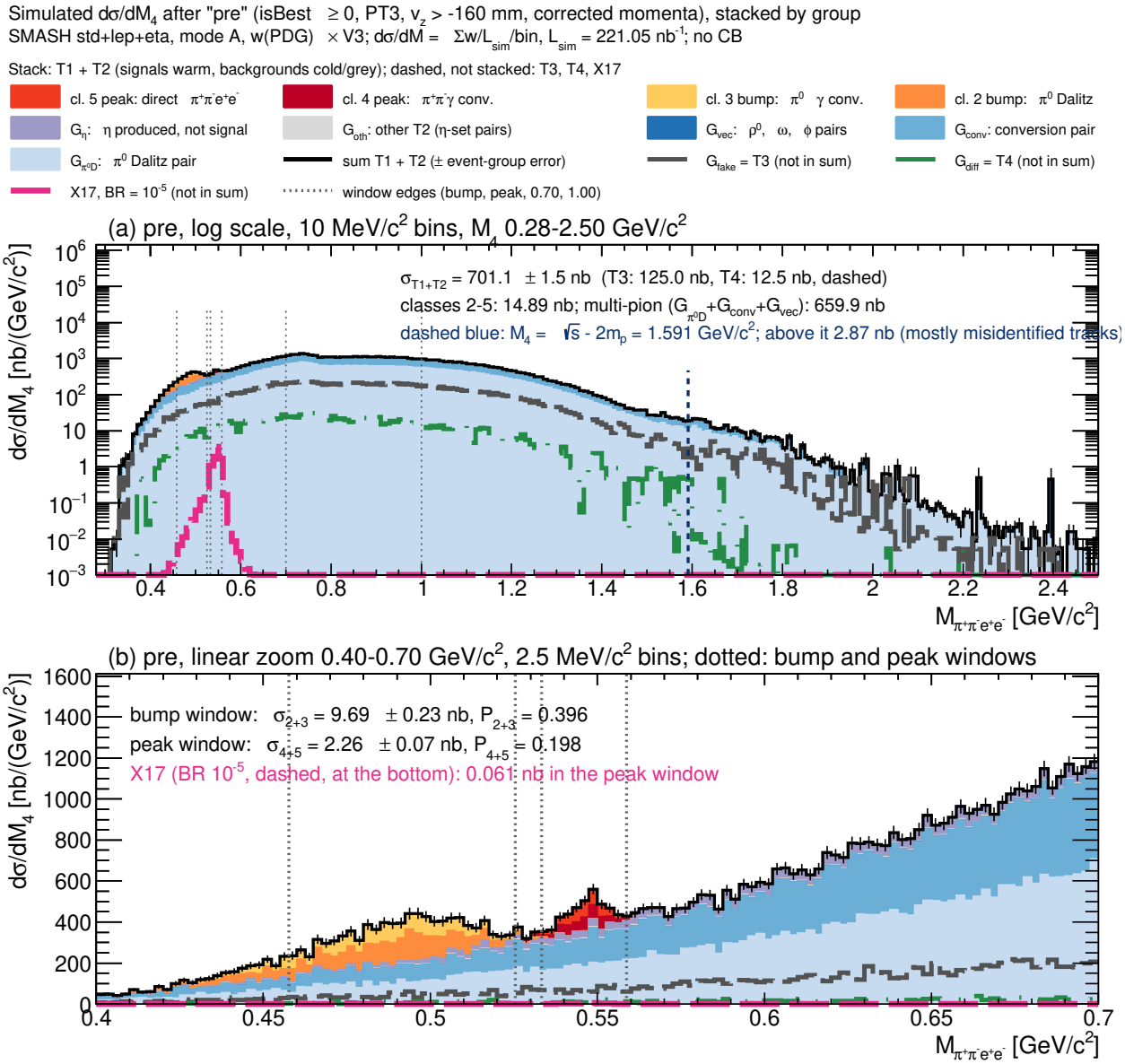


Figure 6: Simulated $d\sigma/dM(\pi^+\pi^-e^+e^-)$ after "pre", stacked by group (signal classes warm, background groups cold or grey). Dashed and not stacked: T3, T4 and X17 at BR $= 10^{-5}$. (a) Log scale, 10 MeV/c^2 bins; the dashed blue line is the kinematic limit. (b) Linear zoom, 2.5 MeV/c^2 bins, with the bump and peak windows (dotted).

The cut sequence. The four η-selection cuts of the analysis are applied one after the other: OA $> 2°$, $\alpha_{\text{lab}} < 50°$, $M(\pi^+\pi^-) < 0.46\,\text{GeV}/c^2$, $\alpha_\eta > 140°$. All four together are the selection "slides". Table 21 gives the cross sections, the purity $P =$ (signal in the window)/(T1+T2 in the window) and $F = S/\sqrt{S+B}$, with S the signal rows and B the other T1+T2 rows of the window at L_{data}. Figure 7 shows the spectrum after each step.

Table 21: Cut sequence (nominal order). σ in nb, errors from event groups. P: purity in the window. $F = S/\sqrt{S+B}$ at $L_{\rm data} = 6080.5\,{\rm nb}^{-1}$ (UNCONFIRMED); in brackets F with B + T3 + T4. Columns: the cuts OA$> 2°$, $\alpha_{\rm lab} < 50°$, $M(\pi^+\pi^-) < 0.46\,{\rm GeV}/c^2$ and $\alpha_\eta > 140°$ added one by one; last column: the final selection with $M(\pi^+\pi^-) < 0.42\,{\rm GeV}/c^2$.

	pre	+OA	+$\alpha_{\rm lab}$	+$M_{\pi\pi}$	+α_η	$M_{\pi\pi} < 0.42$
T1+T2, all M_4	701.1 ± 1.5	476.4 ± 1.1	254.0 ± 0.9	89.6 ± 0.5	67.9 ± 0.5	53.0 ± 0.4
classes 2–5, all M_4	14.89 ± 0.27	10.58 ± 0.22	10.10 ± 0.22	10.06 ± 0.22	7.60 ± 0.18	7.26
bump: classes 2+3	9.69 ± 0.23	6.88 ± 0.20	6.61 ± 0.19	6.61 ± 0.19	5.60 ± 0.17	5.59 ± 0.17
bump: T1+T2	24.48 ± 0.33	17.93 ± 0.28	16.71 ± 0.27	16.71 ± 0.27	15.08 ± 0.25	14.84 ± 0.25
bump: P	0.396 ± 0.007	0.383 ± 0.008	0.395 ± 0.008	0.395 ± 0.008	0.371 ± 0.008	0.377 ± 0.008
bump: F	152.7 ± 3.0	126.6 ± 3.0	126.0 ± 3.0	126.0 ± 3.0	112.5 ± 2.9	113.2 ± 2.9
	(143.2)	(119.3)	(119.0)	(119.0)	(106.2)	(106.9)
peak: classes 4+5	2.26 ± 0.07	1.57 ± 0.05	1.40 ± 0.05	1.37 ± 0.05	1.37 ± 0.05	1.07 ± 0.04
peak: T1+T2	11.45 ± 0.18	8.45 ± 0.15	6.97 ± 0.14	6.79 ± 0.13	6.79 ± 0.13	5.64 ± 0.12
peak: P	0.198 ± 0.006	0.185 ± 0.006	0.201 ± 0.007	0.202 ± 0.007	0.202 ± 0.007	0.190 ± 0.007
peak: F	52.1 ± 1.5	42.0 ± 1.3	41.5 ± 1.4	41.0 ± 1.4	41.0 ± 1.4	35.3 ± 1.2
	(48.1)	(39.0)	(38.4)	(38.0)	(38.0)	(32.6)

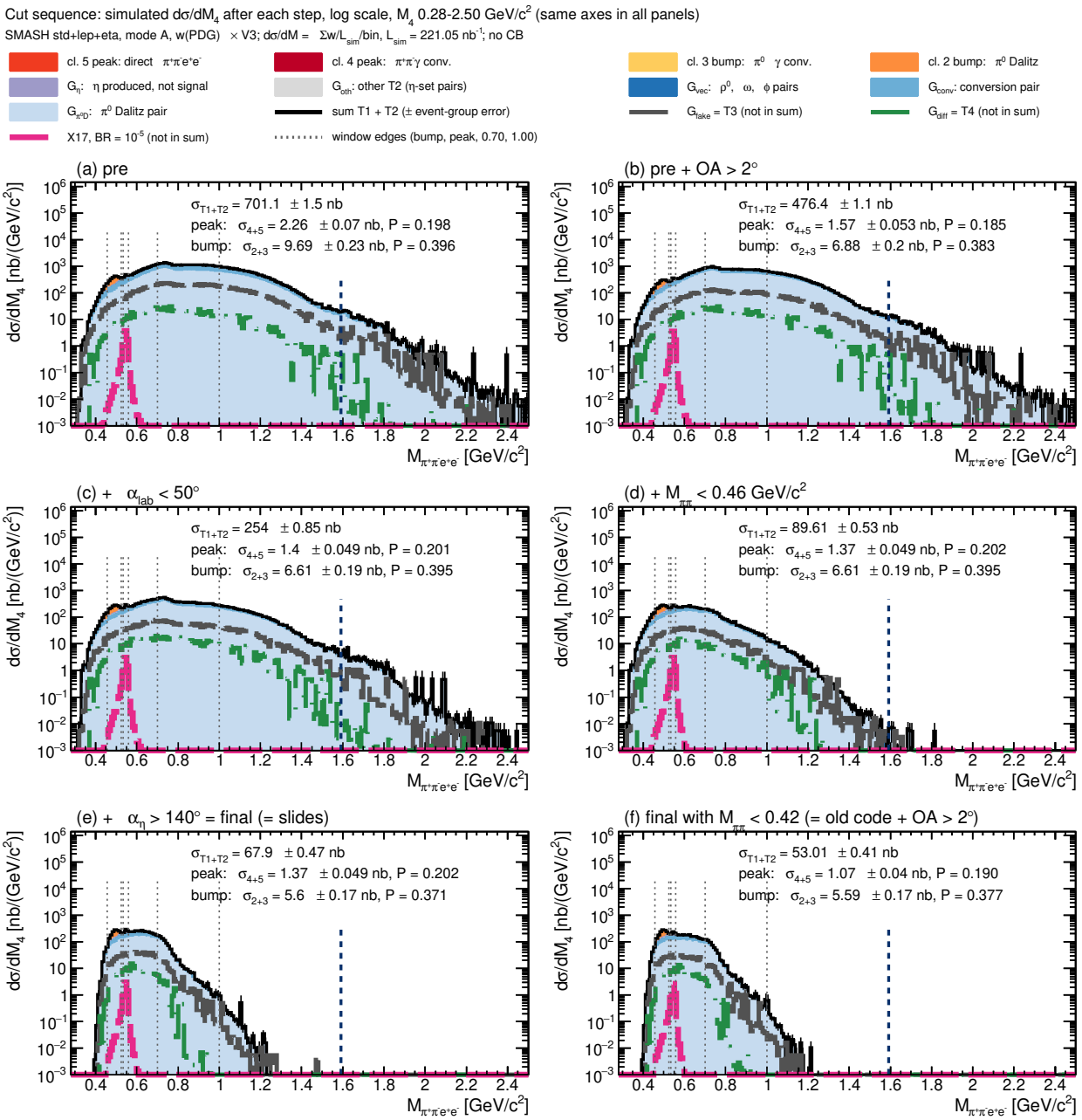


Figure 7: Simulated $d\sigma/dM(\pi^+\pi^-e^+e^-)$ after each step of the cut sequence (same axes in all panels; legend as in Fig. 6). (a) pre; (b)–(e) the four cuts added one by one; (e) is "slides"; (f) the final selection with $M(\pi^+\pi^-) < 0.42\,\mathrm{GeV}/c^2$.

Where each cut acts. Table 22 gives, for each cut applied alone to "pre", the fraction of the background of each $M(\pi^+\pi^-e^+e^-)$ region that it removes.

- OA$> 2°$ removes 25–36 % of the background in every region: it acts everywhere the same.
- α_{lab} and $M(\pi^+\pi^-)$ act mainly above the η: $M(\pi^+\pi^-) < 0.46$ removes nothing in the bump window, because there $M(\pi^+\pi^-) \le M(\pi^+\pi^-e^+e^-) - M(e^+e^-)$ is already below the limit, and 1.8 % in the peak window.
- α_η is a band-pass in $M(\pi^+\pi^-e^+e^-)$. It is the angle between $\pi^+\pi^-$ and e^+e^- in a frame obtained by boosting with the measured total momentum and an energy computed with the η mass (`Physics::openingAngleInMassConstrainedRestFrame`). For $M(\pi^+\pi^-e^+e^-) = m_\eta$ this frame is the true rest frame and the angle is exactly 180°. In the peak window the cut

therefore removes nothing (efficiency 1.000 for every group); it removes 75 % of the background below the bump, 78 % in 0.70–1.00 and 99 % above 1.00 GeV/c^2, and 10–14 % of the bump signal in the bump window.

Table 22: Fraction of the background removed by each cut alone (from "pre"). Background: T1+T2 minus the signal of the window in the bump and peak windows, T2 elsewhere. Errors 0.002–0.012 in the first three columns, below 0.002 elsewhere.

cut	< 0.4576	bump	peak	0.56–0.70	0.70–1.00	> 1.00	all $M(\pi^+\pi^-e^+e^-)$ (T2)
OA$> 2°$	0.247	0.252	0.250	0.276	0.317	0.363	0.321
$\alpha_{\text{lab}} < 50°$	0.014	0.077	0.166	0.307	0.473	0.559	0.455
$M(\pi^+\pi^-) < 0.46\,\text{GeV}/c^2$	0.000	0.0005	0.018	0.254	0.631	0.876	0.616
$\alpha_\eta > 140°$	0.751	0.053	0.000	0.075	0.781	0.987	0.702

The order of the cuts matters for the attribution. Inside the two windows the cuts act almost independently: the fraction attributed to a cut changes by 10–20 % (relative) between the nominal and the reverse order. Over the whole spectrum it changes completely: α_η removes 69 % of T1+T2 when applied first, but 3 % when applied last. A statement "this cut removes that much background" is therefore meaningful only per window and together with the order.

The resulting shape. After "slides" the multi-pion background is still the largest part of every window: 0.72 below the bump, 0.55 in the bump window, 0.70 in the peak window, 0.89 in 0.56–0.70 GeV/c^2. T1+T2 above 0.70 GeV/c^2 drops from 545.9 to 9.96 nb, from 78 % to 15 % of the spectrum. Above 1.00 GeV/c^2 only 0.037 nb remain, with $N_{\text{eff}} = 33$.

Main result of the spectrum study. In the simulation the four cuts do not raise the purity of either window (bump 0.396 → 0.371, peak 0.198 → 0.202), and F falls at every step (peak 52.1 → 41.0, bump 152.7 → 112.5). What the cuts do is remove the multi-pion background outside the windows. A gain from these cuts in data can only come from what the simulation does not contain: the CB, or a data/SMASH difference of the multi-pion background. This must be checked with data.

5.2 The signal at generator level

Number of produced decays. The PLUTO generator file of the η set holds the true kinematics of $N = 1\,181\,801$ direct decays from 1000 of the 100 000 files. Assuming these files are representative, the set contains $N_{\text{direct}} = 100\,N = 1.1818 \times 10^8$ direct decays. Every η of the set is forced, half into the direct decay, so the set holds $N_\eta = 2N_{\text{direct}} = 2.364 \times 10^8$ η mesons, i.e. $\sigma(pp \to \eta X) = N_\eta / L_{\text{sim}} = 1.069$ mb in SMASH. The decays of the other channels are N_η times the PDG branching ratio (Table 23). The statistical error of this scale is 0.09 %; whether the 1000 files are representative is UNVERIFIED.

Table 23: Produced decays per set (10^{10} events) and production cross sections, from N_η and PDG branching ratios. Class 3 is counted per $\eta \to \pi^+\pi^-\pi^0$ decay with any π^0 decay (the conversion happens in the detector), class 4 per $\eta \to \pi^+\pi^-\gamma$ decay. Scale error 0.09 % for every row.

Class	Channel	produced decays	σ_{prod} [nb]
2	$\eta \to \pi^+\pi^-\pi^0$, $\pi^0 \to e^+e^-\gamma$	6.360×10^5	2 877.2
3	$\eta \to \pi^+\pi^-\pi^0$ (any π^0 decay)	5.417×10^7	245 075
4	$\eta \to \pi^+\pi^-\gamma$	9.974×10^6	45 123
5	$\eta \to \pi^+\pi^-e^+e^-$	6.334×10^4	286.56
6	$\eta \to \pi^+\pi^-X17$, $X17 \to e^+e^-$, BR $= 10^{-5}$	2.364×10^3	10.693

Shapes (Fig. 8). The physics weights replace the generator distribution of the direct decay by Kroll–Wada × the P-wave rate × the form factors × the decay angles (V3). The $M(e^+e^-)$ spectrum then becomes much harder:

fraction of produced decays with $M(e^+e^-) >$	2	10	17	20	50 MeV/c^2
direct, generator	0.513	0.099	0.057	0.047	0.016
direct, V3	0.868	0.478	0.351	0.313	0.118
bump π^0 Dalitz (Kroll–Wada)	0.860	0.443	0.308	0.268	0.070

The mean $M(\pi^+\pi^-)$ is 0.4007 GeV/c^2 for the direct decay (V3), 0.3623 for the bump (KLOE-2; phase space 0.3465), 0.4057 for the conversion peak ($|F_V|^2$; phase space 0.4014) and 0.3985 for X17 (three-body phase space, ASSUMED). The bump ends at $m_\eta - m_{\pi^0} = 0.413$ GeV/c^2.

Direct decays under the X17 line. The direct decay has a continuous $M(e^+e^-)$ spectrum and contributes under any X17 line. With V3, the produced direct decays within $|M(e^+e^-) - 17| < d$ are 4.04, 7.95, 15.9 and 32.4 nb for $d = 0.5$, 1, 2 and 4 MeV/c^2; X17 with BR $= 10^{-5}$ is 10.69 nb (Fig. 8d).

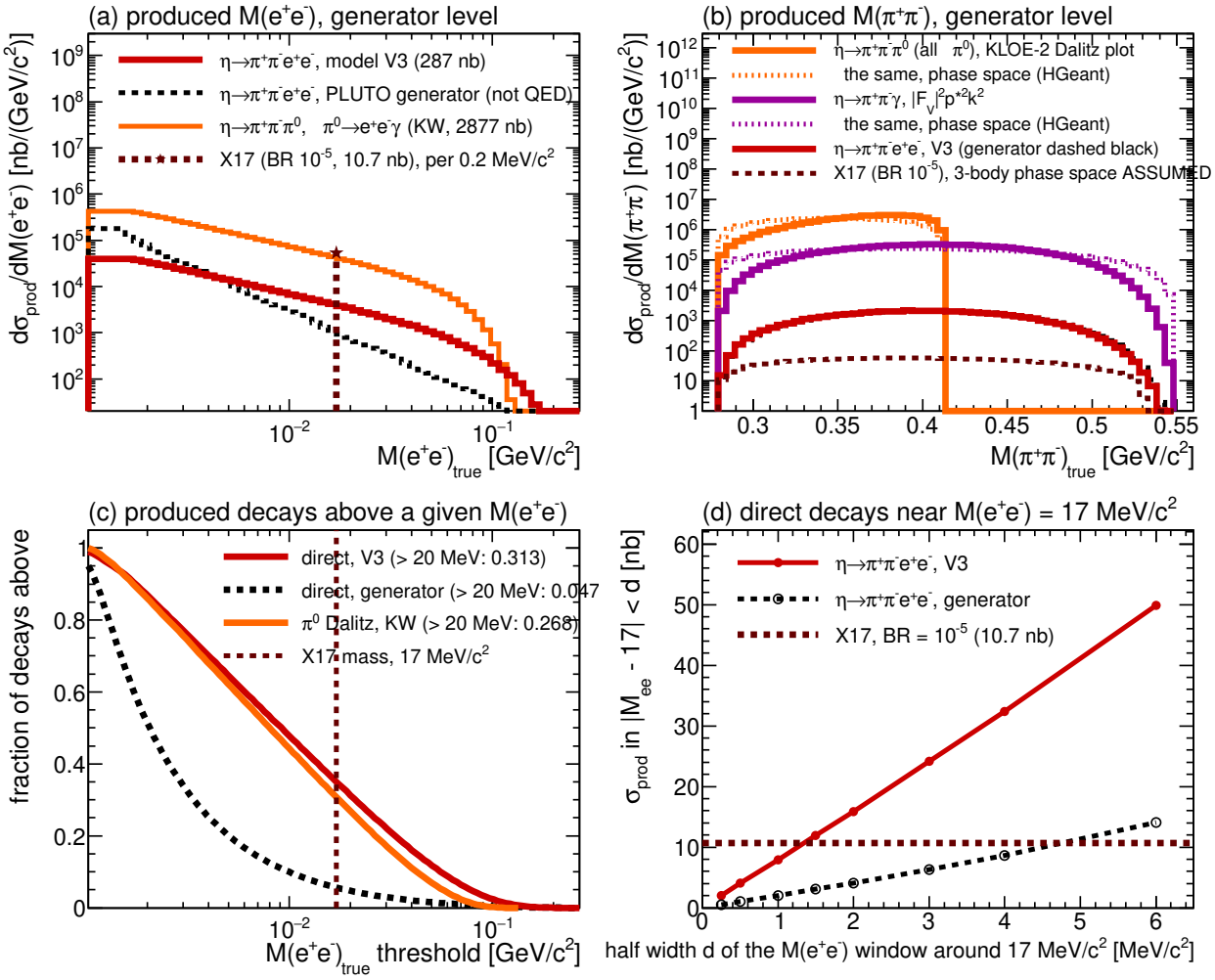


Figure 8: Generator level, absolute scale ($\sigma_{\rm prod}$). (a) $M(e^+e^-)$ of the direct decay (V3 and generator), of the π^0 Dalitz pair of the bump, and the X17 line. (b) $M(\pi^+\pi^-)$ of all channels: physics models (solid) against the phase space of HGeant (dotted). (c) Fraction of decays above a given $M(e^+e^-)$. (d) Direct decays in a window $\pm d$ around 17 MeV/c^2 against X17 at BR $= 10^{-5}$.

5.3 Acceptance and resolution

Acceptance × efficiency ε = (rows of the class after "pre", weighted) / (produced decays). Every accepted decay gives exactly one row of its class. Table 24 gives the totals.

Table 24: Acceptance × efficiency of "pre" per produced decay, in %. Classes 3 and 4 include the conversion probability (per $\eta \to \pi^+\pi^-\pi^0$ and per $\eta \to \pi^+\pi^-\gamma$ decay). X17: at BR = 10^{-5}.

				ε, all M_4		ε (V3), window	
class	rows	N_{eff}	σ_{acc} [nb]	V3	no weights	peak	bump
5 direct	571 246	171 004	1.4168 ± 0.0034	0.4944 ± 0.0013	0.4834	0.4213	0.0249
6 X17	787 659	787 653	0.0713	0.6665 ± 0.0010	0.6665	0.5714	0.0352
2 bump Dalitz	1 649	1 332	7.05 ± 0.19	0.245 ± 0.007	0.275	0.0010	0.194
3 bump conv.	1 187	948	5.19 ± 0.17	0.00212 ± 0.00007	0.00238	0	0.00167
4 peak conv.	307	270	1.24 ± 0.08	0.00275 ± 0.00017	0.00272	0.00234	0.00013

Figure 9 shows ε against the true $M(e^+e^-)$ and $M(\pi^+\pi^-)$.

- **True** $M(e^+e^-)$ (direct decay, V3): 0.34–0.37 % at 2–4 MeV/c^2, 0.49 % at 10, 0.53 % at 16–20, 0.58 % at 45–57, 0.96 % at 100–124 and 1.8–2.0 % above 160 MeV/c^2. A pair of small mass has a small opening angle and is often lost. The two lowest bins are affected by the stored precision of the generator file.
- **Decay angles.** At fixed $M(e^+e^-)$, V3 lowers ε by the factor 0.82 at 5–20 MeV/c^2 and 0.84–0.90 at 20–75 MeV/c^2 (Fig. 9c). The cause is the QED e^+ angle $1 + \cos^2\theta_e^*$: V2, the same model without the angles, changes ε by less than 6 %. Over all $M(e^+e^-)$ V3 still raises ε by 2.3 %, because it moves decays to higher $M(e^+e^-)$.
- **True** $M(\pi^+\pi^-)$**.** ε falls steeply with $M(\pi^+\pi^-)$, because a heavy pion pair leaves slow leptons: direct decay 0.86 % at 0.29–0.30, 0.47 % at 0.41–0.42, 0.016 % at 0.49–0.50 and 0 above 0.52 GeV/c^2.
- **X17 against the direct decay** at the same true $M(e^+e^-)$ (14–20 MeV/c^2): without physics weights the two agree within about 5 % below $M(\pi^+\pi^-) = 0.45$ GeV/c^2 (Fig. 9d). This supports, but does not prove, the phase-space $M(\pi^+\pi^-)$ and the isotropic decay assumed for X17.

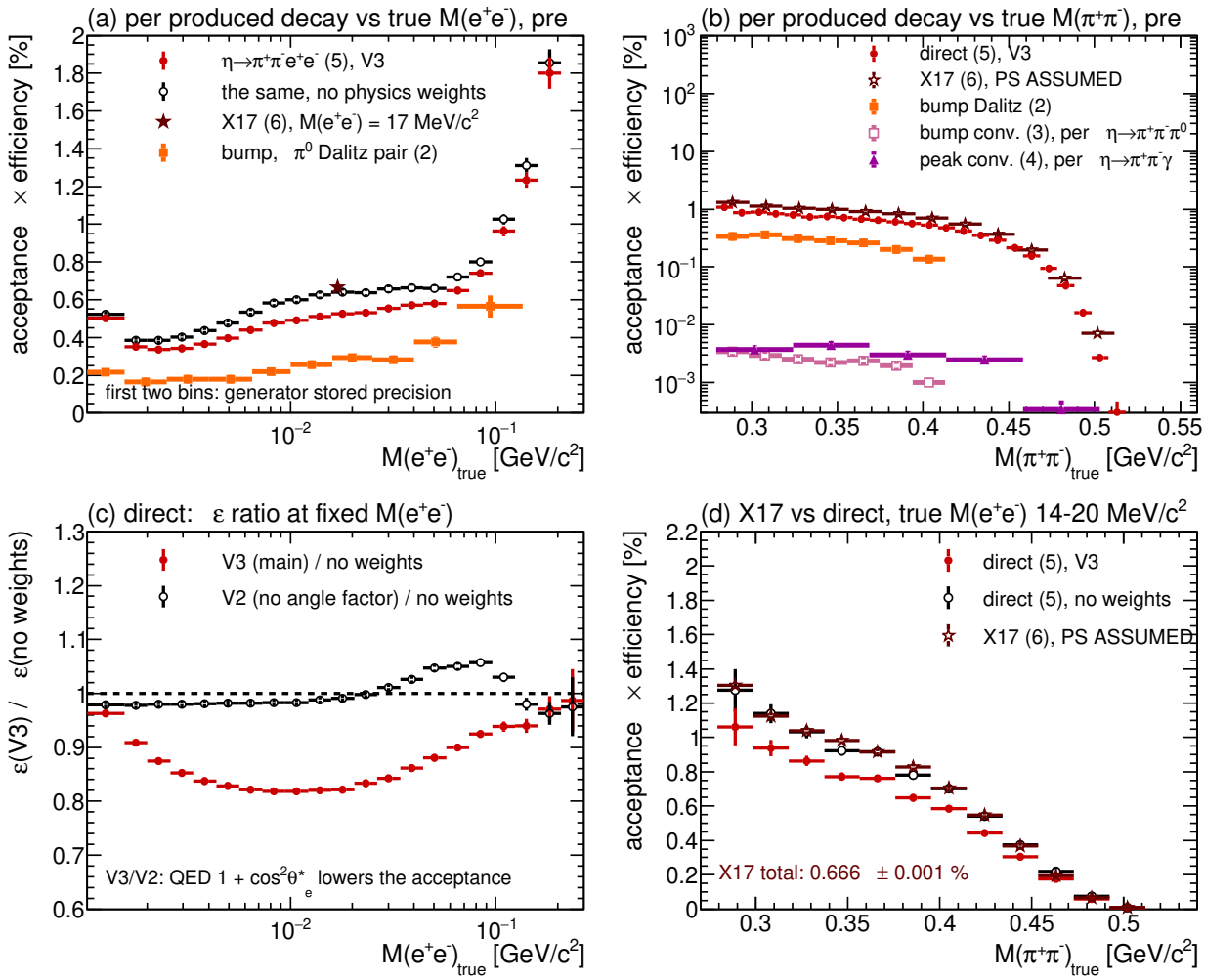


Figure 9: Acceptance × efficiency of "pre" per produced decay. (a) Against the true $M(e^+e^-)$: direct decay (V3 and without physics weights), bump π^0 Dalitz pair, X17. (b) Against the true $M(\pi^+\pi^-)$ for all classes. (c) Ratio V3/no weights and V2/no weights at fixed $M(e^+e^-)$: the effect of the decay angles. (d) X17 against the direct decay at true $M(e^+e^-)$ 14–20 MeV/c^2.

Resolution and scale (Table 25, Fig. 10). The difference reconstructed minus true is fitted with a Gaussian core (iterated in $\pm 2\sigma$); the tails are counted outside $\pm 3\sigma$. The reconstructed $M(\pi^+\pi^-e^+e^-)$ is low by 0.25–0.39 % (median ratio) and has a low-side tail from energy loss and radiation: 6 % of the direct decays lie more than 3σ below the core. The $M(e^+e^-)$ resolution comes from the opening-angle measurement (OA resolution 1.0–1.4°): at 17 MeV/c^2 the core width is 4.21 ± 0.01 MeV/c^2 with a median shift of +0.10 MeV/c^2; 90 % of the X17 rows lie in 9.7–24.3 MeV/c^2.

Table 25: Resolution of $M(\pi^+\pi^-e^+e^-)$ (reconstructed with corrected momenta minus true), after "pre".

class	ΔM_4 [MeV/c^2] median	ΔM_4 [MeV/c^2] core σ	tail fraction below / above 3σ	median $M_4/M_{4,\mathrm{true}}$	$\Delta M_{\pi\pi}$ core σ [MeV/c^2]
5 direct	-2.12 ± 0.02	6.44 ± 0.02	0.057 / 0.014	0.99614	3.49 ± 0.01
6 X17	-2.10 ± 0.01	6.25 ± 0.01	0.063 / 0.013	0.99617	2.98 ± 0.01
2 bump Dalitz	-1.47 ± 0.22	5.47 ± 0.18	0.041 / 0.015	0.99694	3.13 ± 0.12
3 bump conv.	-1.05 ± 0.27	5.68 ± 0.26	0.033 / 0.020	0.99783	3.11 ± 0.13
4 peak conv.	-1.33 ± 0.57	6.13 ± 0.40	0.060 / 0.008	0.99757	3.15 ± 0.26

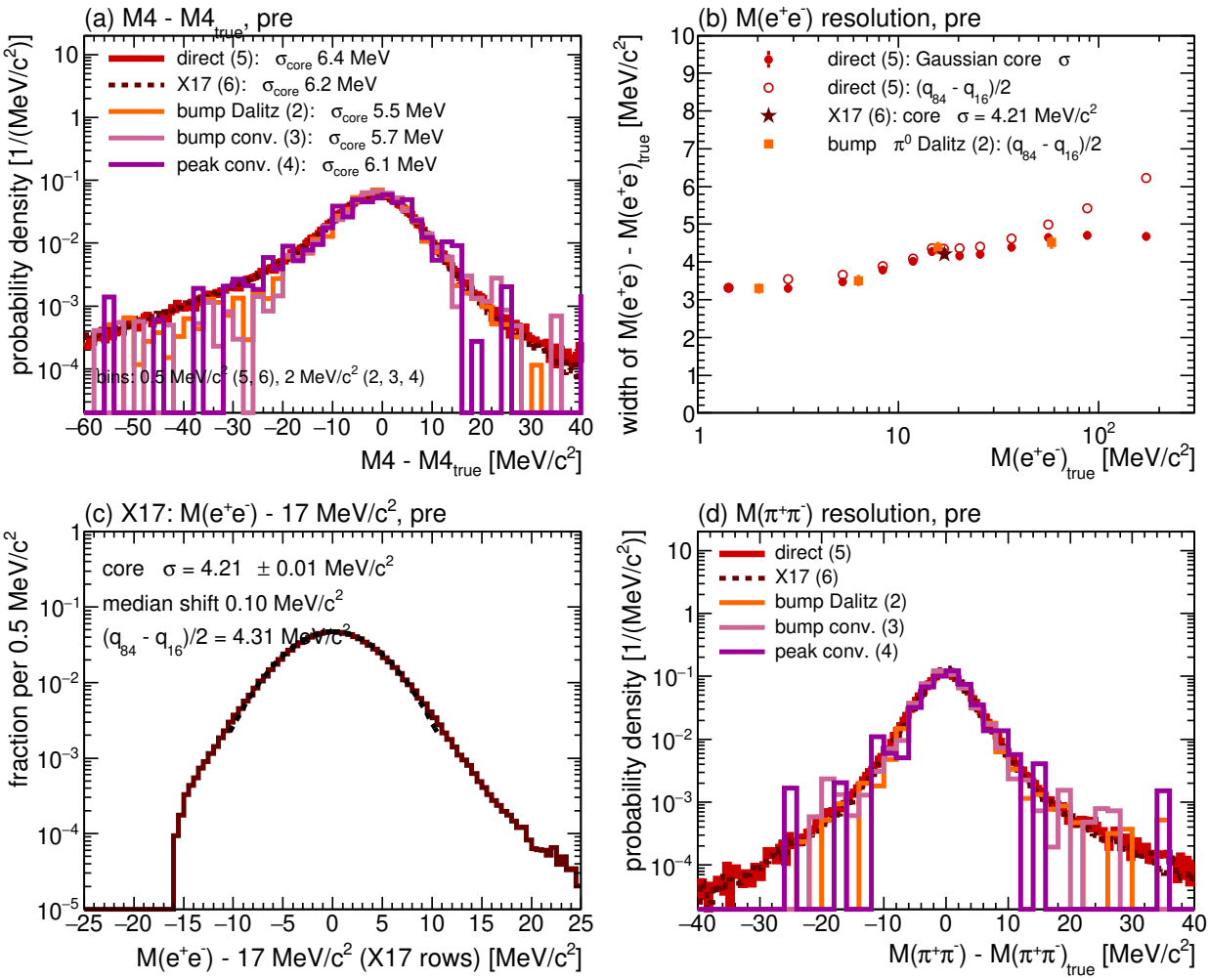


Figure 10: Resolution after "pre". (a) $M(\pi^+\pi^-e^+e^-) - M_{4,\mathrm{true}}$ per class. (b) Width of $M(e^+e^-) - M_{ee,\mathrm{true}}$ against the true $M(e^+e^-)$. (c) Reconstructed $M(e^+e^-)$ of the X17 rows minus 17 MeV/c^2. (d) $M(\pi^+\pi^-) - M_{\pi\pi,\mathrm{true}}$.

Check of the peak window. A Gaussian fit of the simulated peak (classes 4+5, V3, 1 MeV/c^2 bins, iterated in $\pm 2\sigma$) gives $\mu = 0.54637 \pm 0.00020$ GeV/c^2 and $\sigma = 6.09 \pm 0.19$ MeV/c^2, so $\mu \pm 1.9\sigma =$ 0.5348–0.5579 GeV/c^2. The fixed window (0.5333–0.5587) is 1.5 MeV/c^2 wider on the low side. Nine fit choices (bin width, fit range) move the lower edge by +0.1 to +3.2 MeV/c^2; the direct decay alone (N_{eff} 171 000) reproduces the fixed window within 0.7 MeV/c^2. The fixed window is therefore kept. It holds 0.851 of the simulated peak, 2.262 ± 0.070 nb (direct 1.207 ± 0.003, conversion 1.054 ± 0.070).

The opening angle separates conversions only weakly. The true OA of a conversion pair is small (median 0.51°), but the reconstructed OA is larger by about 1.6° (median 2.2°), because of the OA resolution and the need for two separated tracks. The fraction with OA$>2°$ is 0.811 (direct), 0.955 (X17), 0.814 ± 0.011 (bump Dalitz), 0.576 ± 0.016 (bump conversion) and 0.567 ± 0.030 (peak conversion). A cut on $M(e^+e^-)$ separates about as well: $M(e^+e^-) > 15$ MeV/c^2 keeps 0.50 of the direct decay and 0.02 of the conversion peak. OA does not separate X17 from the direct decay at the same reconstructed $M(e^+e^-)$.

5.4 Cut flow per signal class and background composition

Eight cuts are studied: the four of "slides", both pions in the vertex fit, $\max|z_\pi - z_{\mathrm{vertex}}| < 40$ mm (written $|\Delta z_\pi|$), $\mathrm{MM}_4 < 2.85$ GeV/c^2 and $p_{T,4} > 400$ MeV/c. For each cut we compute the efficiency

ε_S of the signal and ε_B of the background T2 in a window, and

$$Q = \varepsilon_S/\sqrt{\varepsilon_B}. \tag{4}$$

Q is the factor by which the cut changes $S/\sqrt{B}$ when $B \gg S$; it needs no luminosity. A cut carries information if $Q > 1 + \max(0.02, 2\delta Q)$ and is harmful if Q is below the mirror value. "Alone" means the cut on "pre"; "N−1" means the cut on top of the other three slides cuts (for the four other cuts: on top of "slides"). Errors are from event groups, with pass and fail rows of one event correlated.

Table 26: Efficiencies and Q of each cut alone (from "pre") in the windows. Peak: S = classes 4+5; bump: S = classes 2+3; X17: S = X17, B = T2 + classes 4+5 (peak window). The error is in brackets, in units of the last digit.

	peak window			bump window			X17 (peak window)	
cut	ε_S	ε_B	Q	ε_S	ε_B	Q	ε_S	Q
OA$> 2°$	0.693(16)	0.749(8)	0.801(19)	0.710(11)	0.748(7)	0.820(13)	0.957	1.114(6)
$\alpha_{\text{lab}} < 50°$	0.916(9)	0.834(7)	1.003(10)	0.970(3)	0.922(4)	1.010(4)	0.892	0.967(3)
$M(\pi^+\pi^-) < 0.46$	0.981(4)	0.982(2)	0.990(4)	1.000	1.000	1.000	0.980	0.989(1)
$\alpha_\eta > 140°$	1.000	1.000	1.000	0.858(10)	0.947(3)	0.882(10)	1.000	1.000
pions in vertex fit	0.938(7)	0.806(7)	1.045(9)	0.923(6)	0.784(6)	1.042(8)	0.931	1.020(4)
$\lvert\Delta z_\pi\rvert < 40\,\text{mm}$	0.963(5)	0.829(7)	1.058(7)	0.944(6)	0.813(6)	1.047(7)	0.962	1.040(3)
$\text{MM}_4 < 2.85$	0.963(5)	0.904(6)	1.013(7)	0.949(5)	0.886(5)	1.008(6)	0.963	1.007(3)
$p_{T,4} > 400\,\text{MeV}/c$	0.795(12)	0.588(9)	1.037(18)	0.849(8)	0.712(7)	1.006(11)	0.791	0.998(6)
slides (all four), from pre	0.606(16)		0.790(22)	0.578(12)		0.722(15)		1.077(7)

What the numbers say (Table 26, Figs. 11 and 12):

- **Inside the windows the cuts hardly separate signal from the simulated background.** Only the two pion-vertex cuts have $Q > 1$ significantly ($Q = 1.04$–1.06 in both windows). They select the same rows: given all other cuts, each adds nothing ($Q = 0.99$–1.02). Signal pions come from the primary vertex; 9–15 % of the background pion pairs contain a $\pi \to \mu$ decay and 28–32 % secondary pions.

- **OA$> 2°$ is harmful for the η windows** ($Q = 0.80$ peak, 0.82 bump): it removes the signal conversions (classes 3 and 4 keep 0.56–0.57) at the same rate as the background conversions. It is informative only for X17 ($Q = 1.114 \pm 0.006$), whose pairs have a median OA of 4.6° against 2.1–2.2° for conversions.

- $\alpha_\eta > 140°$ does nothing in the peak window and costs bump signal ($Q = 0.882 \pm 0.010$): it keeps 0.86 of the bump and 0.95 of T2.

- **The slides cuts are an $M(\pi^+\pi^-e^+e^-)$ band-pass** (Fig. 11): T2 kept by "slides" is 0.17 below the bump, 0.64 in the bump window, 0.59 in the peak window, 0.30 in 0.56–0.70, 0.029 in 0.70–1.00 and 1.8×10^{-4} above $1.00\,\text{GeV}/c^2$. Inside the windows they cost 39 % (peak) and 42 % (bump) of the signal. Their value is a cleaner spectrum around the η (fits, side bands) and, in data, possibly a smaller CB; neither can be measured in this simulation.

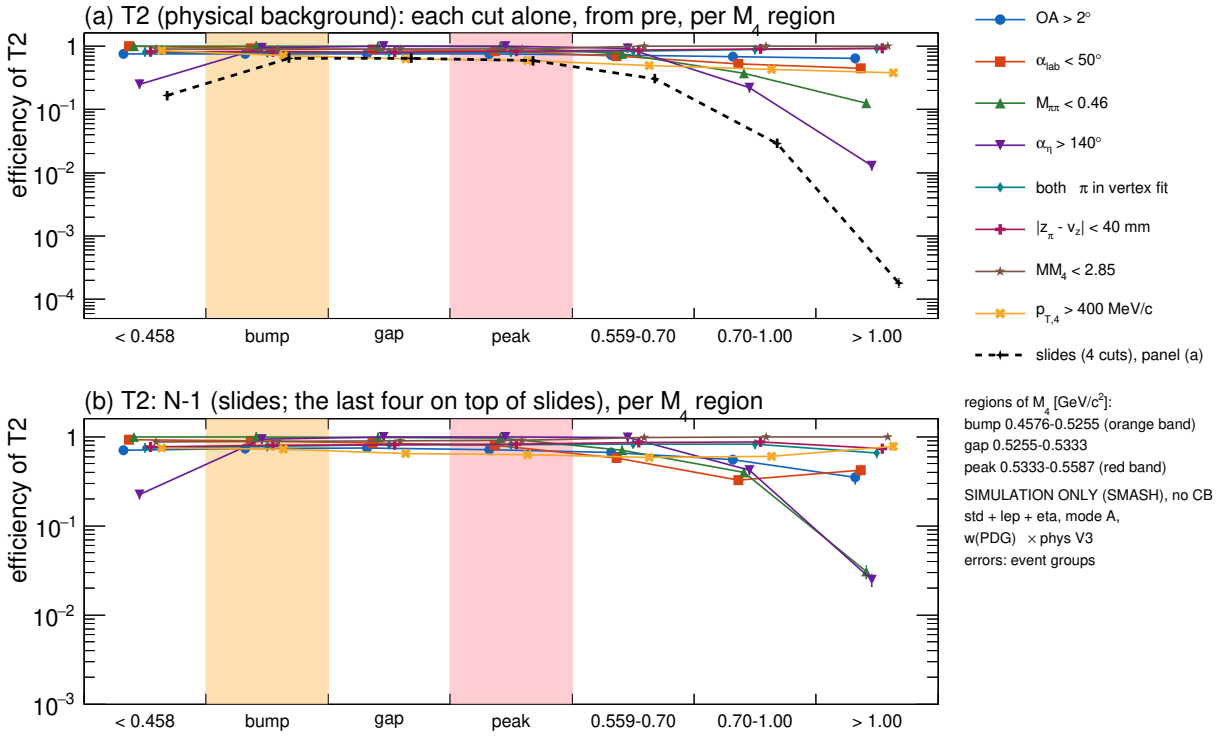


Figure 11: Efficiency of the physical background T2 per $M(\pi^+\pi^-e^+e^-)$ region. (a) Each cut alone on "pre", and the four slides cuts together (dashed). (b) N−1: each cut on top of the other slides cuts (the last four cuts on top of "slides"). Errors from event groups.

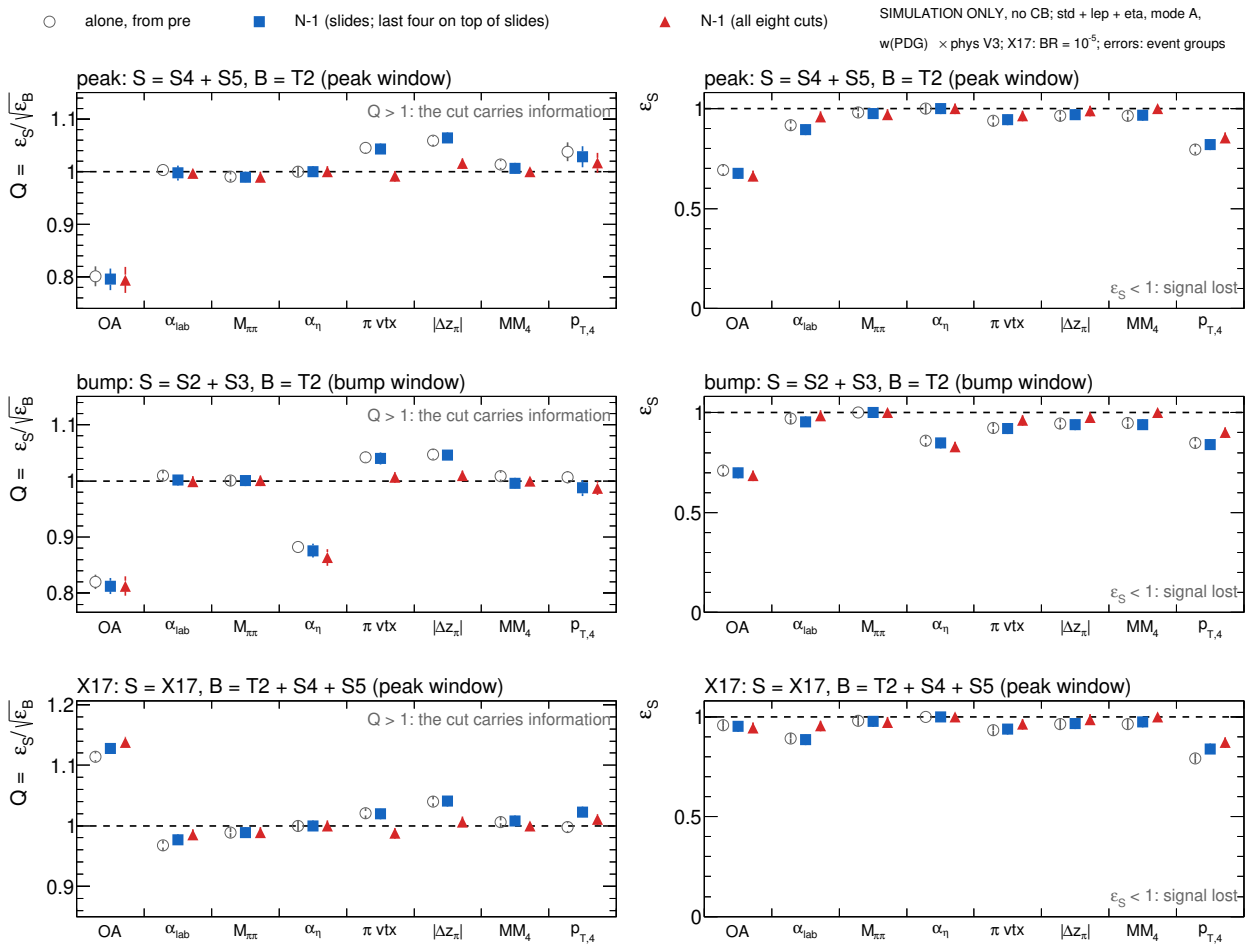


Figure 12: $Q = \varepsilon_S/\sqrt{\varepsilon_B}$ (left) and ε_S (right) of the eight cuts for the peak, the bump and X17 (BR $= 10^{-5}$): alone (open circles), N−1 with respect to "slides" (squares), N−1 with respect to all eight cuts (triangles).

Background composition (Table 27). The physical background has the same make-up in every $M(\pi^+\pi^-e^+e^-)$ region: π^0 Dalitz pairs (L1, 53–60 % of T2) and conversion pairs (L3, 28–39 %). The pions are mostly both primary (P7, 40–72 %) or mixed/secondary (P8, 13–33 %); 9–15 % of the pion pairs contain a $\pi \to \mu$ decay (P1). G_η is 7–10 % of T2 around the η and $G_{\rm vec}$ is negligible. T3 (not in the sum) is 12–20 % of T1+T2 near the η; 60 % of T3 in the peak window are conversion pairs.

Table 27: Composition after "pre" per $M(\pi^+\pi^-e^+e^-)$ region: σ of T1+T2, share of the signal classes in T1+T2, shares of the groups in T2, and T3, T4 relative to T1+T2 (not in the sum).

region [GeV/c^2]	T1+T2 [nb]	signal share	G_η	$G_{\pi^0\rm D}$	$G_{\rm conv}$	T3	T4
< 0.4576	7.66 ± 0.17	0.310(11)	0.067	0.539	0.384	0.147	0.012
bump	24.48 ± 0.33	0.401(7)	0.082	0.539	0.365	0.119	0.019
gap	2.67 ± 0.09	0.100(10)	0.078	0.535	0.369	0.202	0.028
peak	11.45 ± 0.18	0.201(6)	0.093	0.527	0.364	0.150	0.025
0.5587–0.70	109.0 ± 0.6	0.0013(2)	0.064	0.554	0.377	0.186	0.024
0.70–1.00	340.9 ± 1.0	$< 10^{-4}$	0.033	0.573	0.392	0.179	0.018
> 1.00	205.0 ± 0.7	0	0.017	0.595	0.387	0.183	0.014

Statistics. The std-set signal classes limit every number of this part. In the windows after "pre" class 2 has $N_{\rm eff} = 1022$, class 3 718 and class 4 225; after all eight cuts 526, 268 and 79. $G_{\rm vec}$ and

G_{diff} have N_{eff} below 150.

5.5 Optimisation for the η peak and bump

Targets. PEAK: classes 4+5 in the peak window. DIRECT: class 5 in the peak window, with class 4 counted as background. BUMP: classes 2+3 in the bump window. B = all other T1+T2 rows of the window; a variant adds T3 and T4.

Figure of merit and method. $F = S/\sqrt{S+B}$ with S, B in expected rows at L_{data} (UNCONFIRMED; $F \propto \sqrt{L_{\text{data}}}$, the position of the optimum does not depend on it). Nine cut variables are scanned on grids that contain every named selection: OA> (10 values), $\alpha_{\text{lab}} <$ (10), $M(\pi^+\pi^-) >$ (9), $M(\pi^+\pi^-) <$ (10), $\alpha_\eta >$ (8), pions in the vertex fit (off/on), $|\Delta z_\pi| < 40\,\text{mm}$ (off/on), $\text{MM}_4 <$ (8) and $p_{T,4} >$ (6), 13 824 000 combinations in all. Each window row gets a grid index per variable; the weights are filled into a 9-dimensional histogram and summed cumulatively along every axis. After that each cell holds the exact S and B of one combination, and every cell is evaluated (exhaustive search, no random sampling). The table reproduces the engine sums for all named selections and direct row loops for 30 random cells (relative difference $< 10^{-9}$; 378 checks).

Freeze rule (declared before the search). A candidate must reach $F \geq 0.95\,F_{\text{best}}$ on the full sample and on each half (relative to the best of that half). Among the candidates take the one with the fewest active cuts, then the one whose worst grid neighbour (each active threshold moved by one step) has the highest F.

Table 28: Optimisation of the η selection (full sample, $L_{\text{data}} = 6080.5\,\text{nb}^{-1}$ UNCONFIRMED; errors: simulation statistics). $|\Delta z_\pi|$ stands for $\max|z_\pi - z_{\text{vertex}}| < 40\,\text{mm}$. The last two rows are differences of F with the event-group error computed on the difference itself: the same rows enter both selections, so this error is much smaller than the errors of the two F.

	PEAK	DIRECT	BUMP
F at "pre" (purity)	52.12 ± 1.51 (0.198)	27.83 ± 0.23 (0.105)	152.7 ± 3.0 (0.396)
best cell	$M(\pi^+\pi^-)$ 0.32–0.48, $\alpha_\eta > 160°$, $\|\Delta z_\pi\|$, $\text{MM}_4 < 2.85$	$M(\pi^+\pi^-)$ 0.30–0.53, $\alpha_\eta > 150°$, $\|\Delta z_\pi\|$, $\text{MM}_4 < 2.85$	$\alpha_{\text{lab}} < 70°$, $M(\pi^+\pi^-)$ 0.30–0.41, $\|\Delta z_\pi\|$
F of the best cell (purity)	56.18 ± 1.74 (0.262)	29.50 ± 0.28 (0.130)	157.0 ± 3.2 (0.467)
frozen cut	$\|\Delta z_\pi\| < 40\,\text{mm}$, $\text{MM}_4 < 2.85$	$\|\Delta z_\pi\| < 40\,\text{mm}$	none (= pre)
F frozen (F/F_{best})	54.79 ± 1.63 (0.975)	28.95 ± 0.26 (0.982)	152.7 ± 3.0 (0.973)
purity, efficiency (frozen)	0.234 ± 0.007, 0.931 ± 0.007	0.119 ± 0.002, 0.963 ± 0.001	0.396 ± 0.007, 1
F(frozen) − F(pre)	$+2.67 \pm 0.44$	$+1.12 \pm 0.10$	0
F(frozen) − F(pre + pions in vertex fit)	$+1.19 \pm 0.48$	$+0.58 \pm 0.09$	-1.11 ± 0.88

Results (Table 28, Figs. 13 and 14).

- **The surface of F is flat.** The plateau ($F \geq 0.95\,F_{\text{best}}$) contains, for PEAK: OA none or $> 1°$; $\alpha_{\text{lab}} < 40°$ or looser; $M(\pi^+\pi^-) >$ none to 0.34; $M(\pi^+\pi^-) < 0.46$ or looser; any α_η up to 170°; $\text{MM}_4 < 2.7$ or looser; $p_{T,4} > 400$ or looser. The other targets are similar.
- **The frozen cuts are better than "pre", but only a little.** With the correlated error the gain is significant in the simulation: $+2.67 \pm 0.44$ (5.1 %) for PEAK and $+1.12 \pm 0.10$ (4.0 %) for DIRECT. Compared with "pre + pions in vertex fit" the frozen cuts gain $+1.19 \pm 0.48$ (PEAK) and $+0.58 \pm 0.09$ (DIRECT). For BUMP no cut is frozen; the best cell gains $+4.3 \pm 1.0$, but this includes the optimisation bias (below).

- **The PEAK choice was a knife edge.** "$|\Delta z_\pi| < 40\,\mathrm{mm}$" alone reaches 0.966 of F_{best} on the full sample but 0.9494 on half B: it missed the half condition by 0.0006, and the rule added $\mathrm{MM}_4 < 2.85$ (+1 % in F). On random halves the rule picks "$|\Delta z_\pi| < 40\,\mathrm{mm}$" alone in 167 of 200 cases.
- **OA$> 2^\circ$ is off the plateau for all three targets**, even for DIRECT, where it keeps only 0.81 of the direct decay. Only "pions in vertex fit", "peak nominal" and (for BUMP) "bump nominal" are on the plateau (Table 29).
- **Purity.** The cuts can raise the purity at a small loss of F: for BUMP, "bump nominal" gives 0.456 and the best purity on the plateau ($\alpha_{\mathrm{lab}} < 40^\circ$, $M(\pi^+\pi^-) > 0.32$, $|\Delta z_\pi|$) 0.496, against 0.396 at "pre". These purity choices are not stable between halves (the most frequent half pick occurs in only 32–61 of 200 cases).

Table 29: Named selections: F/F_{best} per target (errors about 3 % PEAK, 1 % DIRECT, 2 % BUMP; L_{data} UNCONFIRMED).

selection	PEAK	DIRECT	BUMP
pre	0.928	0.943	0.973
pre + OA$> 2^\circ$	0.748	0.889	0.806
pre + pions in vertex fit	0.954	0.962	0.980
slides	0.730	0.889	0.716
old code	0.784	0.848	0.877
peak nominal	0.941	0.953	0.979
bump nominal	0.732	0.783	0.977

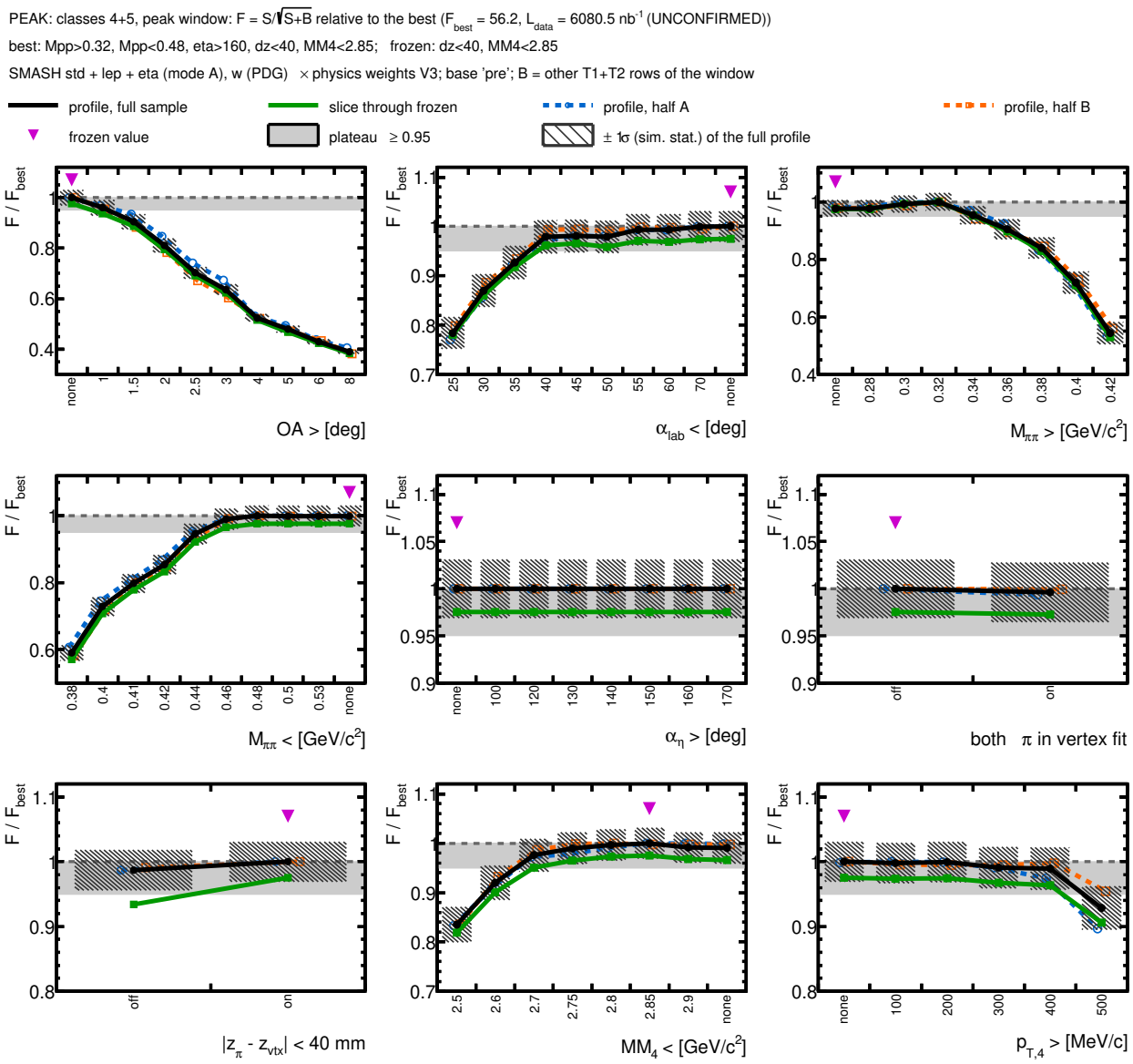


Figure 13: PEAK target: F/F_{best} against each cut variable. Profile = the best F over all other variables at a fixed value (full sample, black; half A and half B relative to their own best, dashed). The hatched band is the $\pm 1\sigma$ statistical error of the full-sample F at each profile point; differences between neighbouring points are much more precise, because the rows are shared. Green: slice through the frozen point. Grey band: plateau (≥ 0.95). Triangle: frozen value. $L_{\text{data}} = 6080.5\,\text{nb}^{-1}$ (UNCONFIRMED).

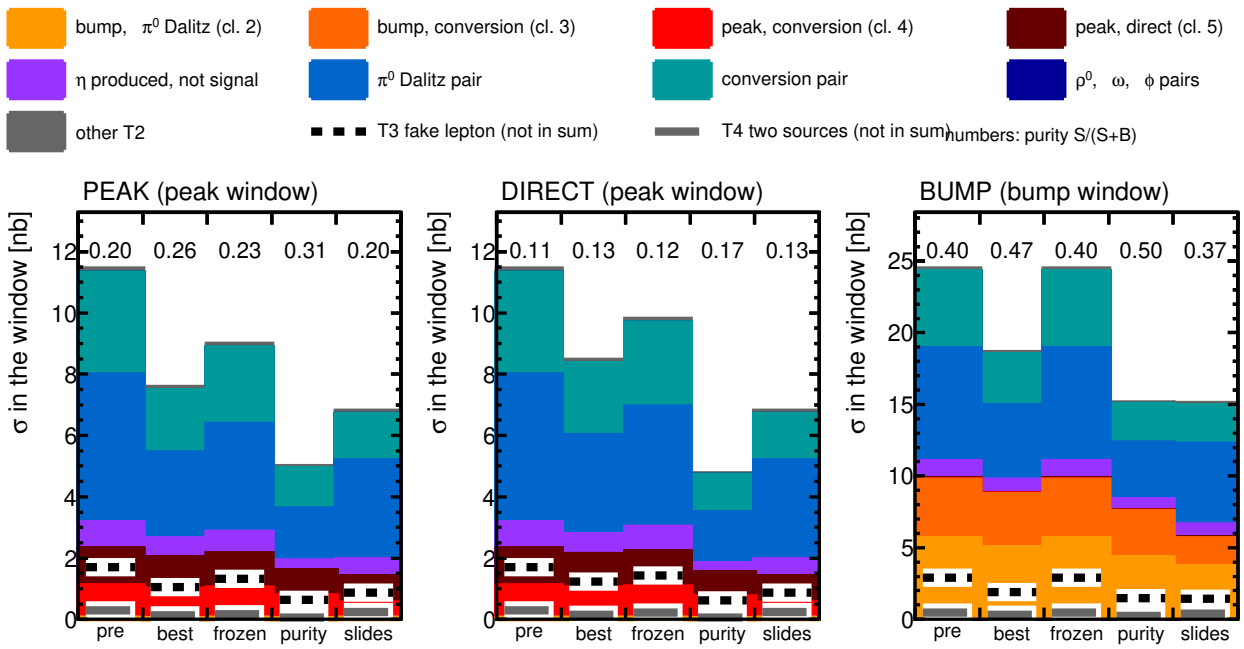


Figure 14: Content of the target window by group, σ in nb, for "pre", the best cell, the frozen cut, the best purity on the plateau and "slides" (stacked: T1+T2 groups). T3 (dashed line) and T4 (dash-dotted line) are drawn at their own level and are not in the stack. Numbers above the bars: purity $S/(S+B)$.

Split-half check of the optimisation bias. The rows are split by event group into two halves (half A = even group number, half B = odd; each half weighted by 2). The search is run on one half and the chosen cell is evaluated on the other. The bias is F(optimising half) − F(other half), averaged over both directions, because a single direction also contains the difference between the halves at fixed cuts (PEAK at "pre": $F_A = 51.28$, $F_B = 52.97$).

	100 random splits (mean ± error of the mean)			deterministic split (A/B)		
target	best cell	freeze rule	purity rule	best cell	freeze rule	purity rule
PEAK	$+2.71 \pm 0.11$ %	$+0.68 \pm 0.09$ %	$+1.55 \pm 0.15$ %	+2.08 %	+1.77 %	+0.61 %
DIRECT	$+0.75 \pm 0.05$ %	$+0.06 \pm 0.01$ %	$+0.80 \pm 0.07$ %	+0.37 %	0.00 %	+1.65 %
BUMP	$+1.10 \pm 0.07$ %	$+0.08 \pm 0.01$ %	$+1.64 \pm 0.09$ %	+0.65 %	0.00 %	+1.30 %

The gain of the best cell over "pre" (PEAK +7.8 %, DIRECT +6.0 %, BUMP +2.8 %) is partly this bias; the freeze rule reduces it to below 1 %. Single directions range from −7.7 to +9.2 %: this is the noise between halves, set by the std set. The limiting components are class 4 (N_{eff} 209 at the frozen PEAK cut; it gives almost all of the 3.0 % error of F), the std part of B (N_{eff} 924) and, for BUMP, the signal itself (N_{eff} 1739).

Frozen η selections (`plots/eta_sim/tables/s5_opt_eta_frozen.txt`): PEAK = pre + $|\Delta z_\pi| < 40$ mm + $\text{MM}_4 < 2.85\,\text{GeV}/c^2$; DIRECT = pre + $|\Delta z_\pi| < 40$ mm; BUMP = pre. **Caveat for data:** $|\Delta z_\pi|$ needs the z of the pion tracks, which the compact data rows (`ScanEngine.h`) do not have yet. The data-ready alternative on the plateau is "pre + pions in vertex fit" (F/F_{best} = 0.954, 0.962, 0.980). Without CB the simulation-only optimum may be looser than the data optimum, since the CB and its subtraction add to the variance and the cuts also reduce the CB.

5.6 A selection aimed at X17 and its expected limit

The X17 line. In the peak window after "pre", the reconstructed $M(e^+e^-)$ of the X17 rows has a mean of $17.25 \pm 0.01\,\text{MeV}/c^2$ and a Gaussian core of $\sigma = 4.04 \pm 0.01\,\text{MeV}/c^2$; 93.3 % lie within

$\pm 2\sigma$. The width depends on the opening angle: 4.5 MeV/c^2 for OA$< 2°$, 3.3 MeV/c^2 for OA$> 8°$; the mean moves from 13.6 (OA$< 3°$) to 20.3 MeV/c^2 (OA$> 8°$), so an OA cut reshapes the line. In $M(\pi^+\pi^-e^+e^-)$ the X17 rows have mean 546.6 MeV/c^2 and $\sigma = 5.9$ MeV/c^2; 85.7 % of them lie in the peak window.

Figure of merit. The expected 90 % CL upper limit on BR $\equiv$ BR$(\eta \to \pi^+\pi^- X17)$ BR$(X17 \to e^+e^-)$ from counting in a window, for BR $= 0$:

$$\mathrm{UL} = \sqrt{\mathrm{UL}_{\mathrm{stat}}^2 + \left(1.645\,\sigma_{B,\mathrm{MC}}/s_1\right)^2}, \qquad \mathrm{UL}_{\mathrm{stat}} = s_{\mathrm{up}}(B)/s_1. \tag{5}$$

Here s_1 is the number of selected X17 rows per unit BR and B the background rows (T1 without X17, plus T2), both at L_{data}; $s_{\mathrm{up}}(B)$ is the CLs upper limit on the signal for an observed count equal to B (continuous Poisson, about $1.645\sqrt{B}$ for large B); and $\sigma_{B,\mathrm{MC}}$ is the simulation (event-group) error of B. The second term is there because a counting analysis takes B from this simulation. $\mathrm{UL}_{\mathrm{stat}}$ alone is the limit with the background known exactly: a statistical limit only.

Two versions. The first optimisation (v1, used by the blind test of Sect. 5.7) used $\mathrm{UL}_{\mathrm{stat}}$ and also optimised the edges of the $M(\pi^+\pi^-e^+e^-)$ window. A review found two problems: the MC error of B (± 307 rows) is larger than the statistical term, and the $M(\pi^+\pi^-e^+e^-)$ window was to stay fixed for all parts of the study. The optimisation was therefore repeated (v2) with Eq. (5) and the fixed peak window. Method: coordinate descent over $M(e^+e^-)$ window, OA (lower and upper), α_{lab}, $M(\pi^+\pi^-)$ (lower and upper), α_η and pions in the vertex fit; in each step the loosest value within 2 % of the step minimum is taken. Two different starting points end at the same optimum.

Table 30: X17 selections after "pre". Rows at $L_{\mathrm{data}} = 6080.5\,\mathrm{nb}^{-1}$ (UNCONFIRMED). UL in units of 10^{-6}; "shape": binned likelihood of $M(e^+e^-)$ 2–62 MeV/c^2 in the $M(\pi^+\pi^-e^+e^-)$ window with a Gaussian X17 line (background shape from the simulation, normalisation known or free). Z: Asimov significance for BR $= 10^{-5}$ with B known.

	v1 (blind test)	v2 (frozen)
cuts in addition to "pre"	$M(\pi^+\pi^-e^+e^-)$ 0.5330–0.5555, $M(e^+e^-)$ 13.0–26.5, pions in vertex fit	$M(\pi^+\pi^-e^+e^-)$ 0.5333–0.5587, $M(e^+e^-)$ 15.5–24.5, pions in vertex fit
X17 rows per unit BR, s_1	2.727×10^7	2.127×10^7
X17 rows at BR $= 10^{-5}$	272.7	212.7
background B [nb] (N_{eff})	1.333 ± 0.051 (696)	0.878 ± 0.039 (501)
background rows $B \pm \sigma_{B,\mathrm{MC}}$	8108 ± 307	5342 ± 239
$\mathrm{UL}_{\mathrm{stat}}$ (B known exactly)	5.48	5.71
UL, Eq. (5) (with the MC error of B)	19.3	19.3
shape, B known / normalisation free	4.85 / 5.40	4.96 / 5.54
$Z(\mathrm{BR} = 10^{-5})$, B known / with $\sigma_{B,\mathrm{MC}}$	3.01 / 0.85	2.89 / 0.85

Results (Table 30, Fig. 15).

- **The MC error of the background sets the limit.** With the MC term the expected limit is 1.93×10^{-5} for both selections, 3.4–3.5 times $\mathrm{UL}_{\mathrm{stat}}$. The re-optimisation gains nothing: the v1 cuts give 1.933×10^{-5}, the v2 cuts 1.931×10^{-5}. "slides" with the same $M(e^+e^-)$ window gives 2.15×10^{-5}.
- **Where the MC error comes from.** At the v2 optimum the variance of B comes from $G_{\pi^0\mathrm{D}}$ (60 %; lep set, N_{eff} 313), G_{conv} (24 %; 19 std rows, N_{eff} 18) and G_η (12 %; N_{eff} 29). One std row stands for 27.5 expected data rows.
- **Cuts.** The $M(e^+e^-)$ window keeps 0.62 of X17 and 0.094 of B; the pion-vertex cut keeps 0.93 of X17 and 0.82 of B. OA, α_{lab}, $M(\pi^+\pi^-)$ and α_η cuts do not lower the limit: at $M(e^+e^-)$

12–22 MeV/c^2 the OA medians of X17 (4.9°), the direct decay (4.6°) and $G_{\pi^0\mathrm{D}}$ (5.2°) are about the same.

- **Composition of** B (v1): $G_{\pi^0\mathrm{D}}$ 0.56, direct decay (class 5) 0.18, G_{conv} 0.12, G_η 0.08, class 4 0.03. The direct decay is an irreducible background under the line.
- **Split halves** (v2 figure of merit): the optimum of half A gives 28.1 on A and 30.2 on B, that of half B 24.4 and 28.1 (in 10^{-6}; each half has a $\sqrt{2}$ larger MC error). The bias is 7–15 %: with the MC term the figure of merit is itself noisy, and the optimum ($M(e^+e^-)$ 13–27.5 on half A, 15.5–23.5 on half B) is not better determined than in v1.

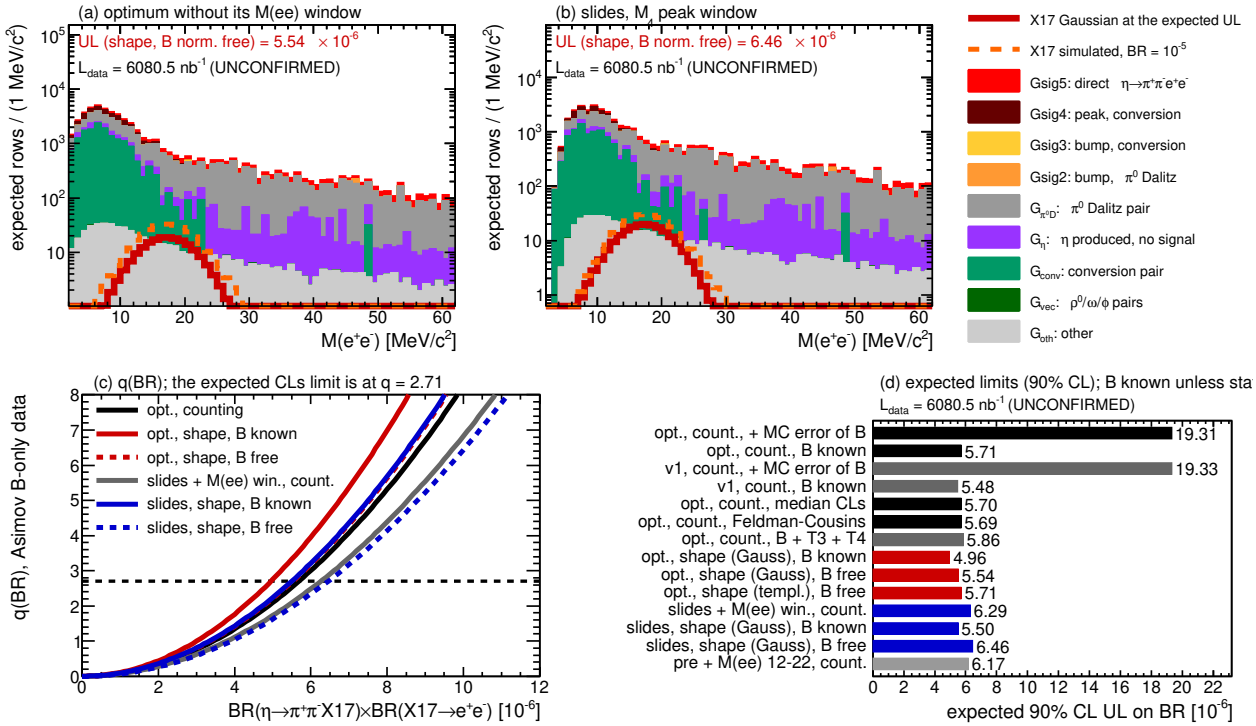


Figure 15: X17 search, expected rows at $L_{\mathrm{data}} = 6080.5\,\mathrm{nb}^{-1}$ (UNCONFIRMED). (a) $M(e^+e^-)$ at the v2 optimum without its $M(e^+e^-)$ window, stacked background groups, X17 as a Gaussian at the expected limit of the shape fit and as the simulated template at BR $= 10^{-5}$. (b) The same for "slides". (c) Test statistic q(BR) on Asimov background-only data; the expected CLs limit is at $q = 2.71$. (d) Expected 90 % CL limits; "B known" ignores the MC error of B.

Frozen X17 selection v2 (`plots/eta_sim/tables/s5_opt_x17_frozen.txt`; v1 kept in `s5_opt_x17_frozen_v1.txt`): pre + peak window + 15.5 ≤ $M(e^+e^-)$ < 24.5 MeV/c^2 + both pions in the vertex fit. Expected 90 % CL limit by counting 1.9×10^{-5} including the MC error of the background; 5.7×10^{-6} if the background were known exactly. These numbers assume $L_{\mathrm{data}} = 6080.5\,\mathrm{nb}^{-1}$ (UNCONFIRMED), no CB, the SMASH background and phase space for X17.

5.7 Blind test of the signal extraction and of the X17 search

The extraction methods planned for the data were tested on simulated pseudo-data whose true content was hidden from the analyst until the procedure was frozen.

Preparation.

1. **Split.** The rows are split by event group: half A (even group number) gives the templates, half B (odd) gives the pseudo-data. Every row of an event goes to the same half.

2. **Templates**: half A with weights $\times 2$, histograms of $M(\pi^+\pi^- e^+ e^-)$ ($1\,\mathrm{MeV}/c^2$ bins) per component and selection (PEAK, DIRECT, BUMP of Sect. 5.5 and the v1 X17 selection), exact window sums, and $M(e^+e^-)$ and $M(\pi^+\pi^- e^+ e^-) \times M(e^+e^-)$ for X17.

3. η **pseudo-data**: 200 toys at $L_{\mathrm{toy}} = L_{\mathrm{sim}}/2 = 110.525\,\mathrm{nb}^{-1}$ from a Poisson bootstrap of half B per event group. An event with largest row weight w_g enters $n \sim \mathrm{Poisson}(w_g)$ times; in each copy one uniform u is drawn and each row with $w_r/w_g > u$ is kept, so that every row has the expected multiplicity w_r. Only T1+T2 rows are used. A check with 4000 toys gave toy variance / predicted variance = 0.95–1.06; the sum of the squared pulls of the variances over the 21 regions was 36.8. Repeated with eight further random seeds, this sum is 9.1–35.7, on average 20.7 ± 10.1 over all nine runs (21 expected; the spread is wider than for 21 independent entries because the selections share rows), and the variance ratio stays within 0.94–1.06: the bootstrap reproduces the variances, the value 36.8 is a fluctuation.

4. **X17 pseudo-data**: 1500 toys at L_{data} (UNCONFIRMED), parametric: in each cell of $M(\pi^+\pi^- e^+ e^-) \times M(e^+e^-)$ signal and background are drawn as Poisson numbers around the half-B expectation ($\times 2 \times L_{\mathrm{data}}/L_{\mathrm{sim}}$). BR $\in \{0, 1, 3, 10, 30\} \times 10^{-6}$, 300 toys each, shuffled, with random ids.

5. **Sealed key**: true rows per toy, group and region; the expectation of half B per toy, called μ, is the quantity a measurement would estimate.

For the weight rules v2 the η and X17 toys were drawn again with the same seed. Because the weights changed, the random draws differ from those of rules v1, so the biases and pulls moved within their statistical errors (by up to 2.3 standard deviations).

Methods (frozen before any yield was computed on the pseudo-data). Peak yield: M1 = Gaussian (fixed at $0.5460\,\mathrm{GeV}/c^2$, $6.7\,\mathrm{MeV}/c^2$) + second-order polynomial, calibrated with the response to the half-A peak template; M2 = template fit (free normalisations of peak, bump and background templates, binned Poisson likelihood with the template variance added); M2b = M2 with the background split into G_η and the rest; M2d = M2 with the peak split into class 5 (free) and class 4 (constrained to its half-A value $\pm$ its MC error); M3 = linear side bands ($7.8\,\mathrm{MeV}/c^2$ on each side) with the signal leakage λ into the side bands corrected, $S = [N_{\mathrm{win}} - (N_L + N_U)/2 \cdot 25.4/7.8]/(1-\lambda)$; M3c = M3 with the curvature of the background taken from half A. Bump yield: B1 = bifurcated Gaussian + η Gaussian + polynomial; B2 = the bump of M2; B3 = edge step in $M(\pi^+\pi^- e^+ e^-)$; B4 = side band 0.5665–0.70 with the background shape from half A. Direct yield: M2d, and D5 = split at $M(e^+e^-) = 10\,\mathrm{MeV}/c^2$ with a 2×2 unfolding. X17: X1 = fit of $M(e^+e^-)$ in the window with free normalisations of the direct decay and the rest; X2 = X1 with the raw half-A template; X3 = X1 on 8–40 MeV/c^2; X4 = X1 with the direct-decay normalisation constrained to 1 ± 0.10; X5 = counting with B from half A. X4 and X5 were added after the X1 results were seen (declared, before their own results). An example toy with the fits is shown in Fig. 16.

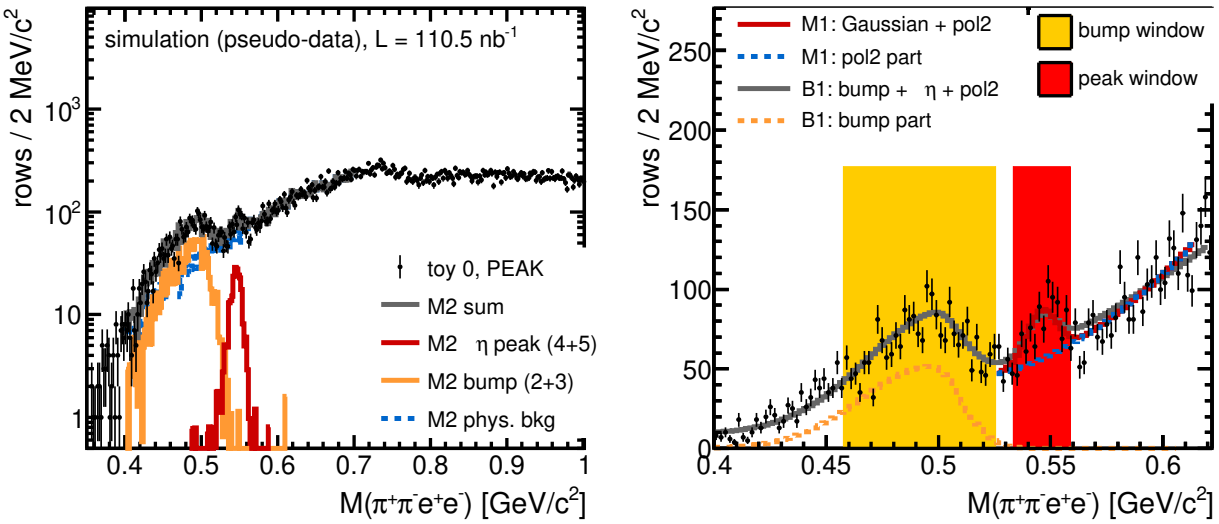


Figure 16: One η toy (pseudo-data, $L = 110.5\,\mathrm{nb}^{-1}$, PEAK selection). Left: $M(\pi^+\pi^-e^+e^-)$ with the components of the template fit M2. Right: zoom with the Gaussian + polynomial fit M1 and the bump fit B1, with the bump and peak windows.

What a bias means here. The templates (half A) and the pseudo-data (half B) are two statistically independent copies of the same model. The truth of half B differs from half A by MC fluctuations: peak (PEAK selection) 241.4 against 224.1 ± 10.3 rows (+7.7 %), bump (BUMP) 1095.6 against 1046.1 ± 35.7 (+4.7 %), class 5 (DIRECT) 128.88 against 128.00 ± 0.48, class 4 (DIRECT) 118.5 against 106.2 ± 10.5 (+11.5 %). All toys share this difference. A method bias can therefore be resolved only beyond the common MC error, which is ±11–15 % for the peak and ±4.5–5 % for the bump.

Table 31: Blind test, η yields (200 toys). Bias = $\langle Y \rangle - \mu$ with the error of the mean over toys and the common MC error (template and parent sample). Pull = $(Y - \mu)/\sigma_{\mathrm{tot}}$ with the error quoted by the analyst (mean/rms). Coverage: fraction of toys with $|Y - \mu| \le \sigma_{\mathrm{tot}}$ (nominal 0.683).

yield, selection (μ)	method	bias [rows]	relative	pull	coverage
peak, PEAK (241.4)	M1 Gauss + pol2	$+7.4 \pm 2.9 \pm 31.3$	+3.1 %	+0.17 / 0.96	0.72
	M2 template fit	$+3.4 \pm 2.0 \pm 27.5$	+1.4 %	+0.08 / 0.80	0.77
	M2b	$+10.4 \pm 2.2 \pm 30.6$	+4.3 %	+0.26 / 0.82	0.74
	M3 side bands	$-51.0 \pm 3.9 \pm 45.3$	−21.1 %	−0.86 / 0.93	0.53
	M3c	$-124.5 \pm 3.9 \pm 56.9$	−51.6 %	−1.82 / 0.80	0.16
bump, BUMP (1095.6)	B4 side band	$+42.3 \pm 3.9 \pm 54.9$	+3.9 %	+0.61 / 0.79	0.66
	M2 template fit	$+84.5 \pm 3.4 \pm 49.7$	+7.7 %	+1.34 / 0.76	0.31
	B1 parametric	$+59.0 \pm 7.1 \pm 78.6$	+5.4 %	+0.56 / 0.96	0.66
	B3 edge step	$+13.8 \pm 13.3 \pm 171.5$	+1.3 %	+0.01 / 0.86	0.76
direct, DIRECT (128.9)	M2d (cl. 4 constr.)	$+26.6 \pm 2.2 \pm 31.0$	+20.6 %	+0.68 / 0.79	0.62
	D5 ($M(e^+e^-)$ split)	$+40.5 \pm 3.9 \pm 39.3$	+31.4 %	+0.70 / 0.96	0.61

η results (Table 31, Fig. 17).

- **Peak.** The template fit M2 and the calibrated Gaussian fit M1 find the peak within +1 to +6 % in all three selections, well inside the common MC error. Per toy they differ by about 12 % (rms 29 rows), so a difference of that size between them in data is normal. M2b is 3–8 % above M2 every time.
- **Linear side bands (M3) lose the peak**, because the background is convex under it and 17 %

of the signal leaks into the side bands. Two independent realisations of the MC noise measure this: the half-B toys give −21 %, −13 %, −16 % and the exact half-A sums (Asimov) −40 %, −43 %, −46 % (PEAK, DIRECT, BUMP); the mean is about -30 ± 14 %. The MC-calibrated version M3c over-corrects (−44 to −52 %), because the curvature from the simulation is itself noisy.

- **Bump.** B4 is closest (+3 to +4 %); the template fit agrees within 3–8 %; the parametric fit B1 comes out 5–10 % high; the edge step B3 is unbiased but has 15–17 % statistical error.
- **Direct decay alone.** $M(\pi^+\pi^-e^+e^-)$ cannot separate class 5 from class 4. M2d gives class 4 = 105.4 (truth 118.5) and therefore class 5 too high by the same amount ($26.6 = 13.5 + 13.1$ rows). The split is limited by the simulation statistics of the conversion peak ($N_{\text{eff}} \approx 100$ in a half).
- **Errors.** The event-group errors describe the spread of the toys: the ratio rms/error is 0.93–1.08 for M1, M3, B1, B3 and D5 and 0.90–1.06 for the template fits and B4, so the errors are adequate. The common MC error is not in the per-toy error and must be added.

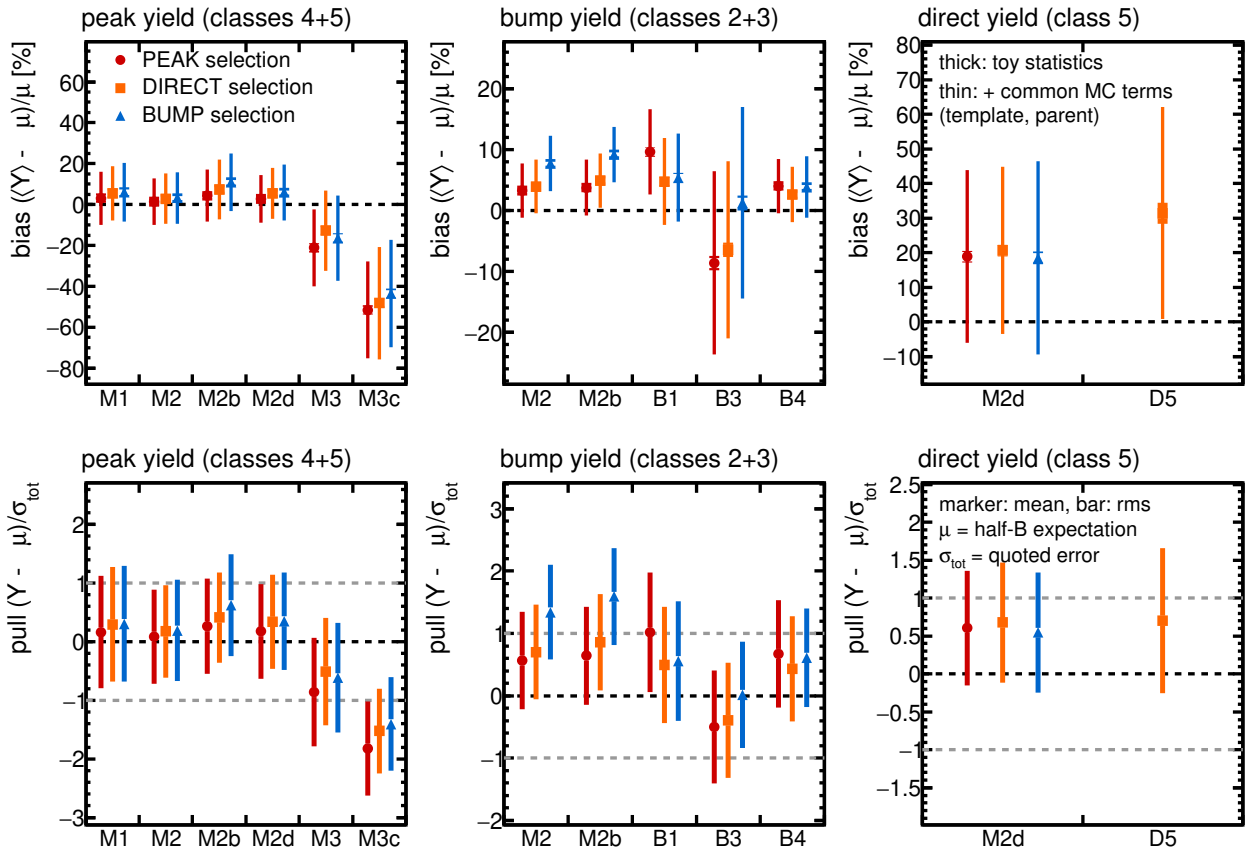


Figure 17: Blind test, η yields. Top: relative bias $(\langle Y\rangle - \mu)/\mu$ per method and selection; thick bars: toy statistics, thin bars: including the common MC terms. Bottom: pull $(Y - \mu)/\sigma_{\text{tot}}$, marker = mean, bar = rms.

X17 results (Table 32, Figs. 18 and 19). Each std row of half B stands for about 55 expected data rows. The background of every X17 toy is therefore the unsmoothed half-B content, with single-bin spikes of 50–200 rows that all toys share and that differ from half A (Fig. 18).

- Every estimator follows the injected BR with a slope of 0.98–1.02.

- The zero point is shifted, by the same amount in all toys: it is set by how well the background under the line is known. Counting with the half-A background (X5) closes exactly: the predicted shift $(B_B - B_A)/s_1 = (7947.8 - 8268.2)/2.725 \times 10^7 = -11.76 \times 10^{-6}$ is measured as $-11.66 \pm 0.19 \times 10^{-6}$ at BR $= 0$.
- Only estimators whose error contains this background uncertainty give valid limits (X1, X4, X5 cover the true BR at every BR). Their median limits at BR $= 0$ are 2.1–6.2 $\times 10^{-5}$, 4–11 times the $\mathrm{UL}_{\mathrm{stat}}$ of v1, and consistent with the 1.9×10^{-5} of Eq. (5).
- Estimators with a small error (X2, X3, X1 with the statistical error only) exclude the true BR. No estimator claimed a 3σ signal at BR $= 0$, but only because the common shift happens to be negative: with the opposite sign X2 and X3 would claim one in 100 % and 98 % of the toys.

Table 32: Blind test, X17 (1500 toys at $L_{\mathrm{data}} = 6080.5\,\mathrm{nb}^{-1}$, UNCONFIRMED; BR in 10^{-6}). Offset: mean extracted BR at injected BR $= 0$. Coverage: fraction of toys whose UL is above the injected BR. X1–X3 were frozen; X4 (X1 with the direct-decay normalisation a_D constrained) and X5 (counting) were added after the freeze.

estimator	offset at 0	slope	median UL at 0 (16–84 %)	UL coverage at 0 / 3 / 10 / 30
X1 window fit	-72.7 ± 0.5	1.004 ± 0.021	31.2 (28.2–34.4)	1.00 / 1.00 / 1.00 / 0.99
X2 raw template	-72.4 ± 0.6	0.977 ± 0.024	2.5 (2.1–3.3)	1.00 / 0.34 / 0.00 / 0.00
X3 8–40 MeV/c^2	-74.9 ± 0.3	0.996 ± 0.011	12.7 (11.8–13.5)	1.00 / 1.00 / 1.00 / 0.00
X4 X1, $a_D = 1 \pm 0.1$	-1.1 ± 0.4	1.016 ± 0.018	61.8 (57.3–67.3)	1.00 at all BR
X5 counting	-11.7 ± 0.2	0.999 ± 0.008	21.1 (19.7–22.7)	1.00 at all BR

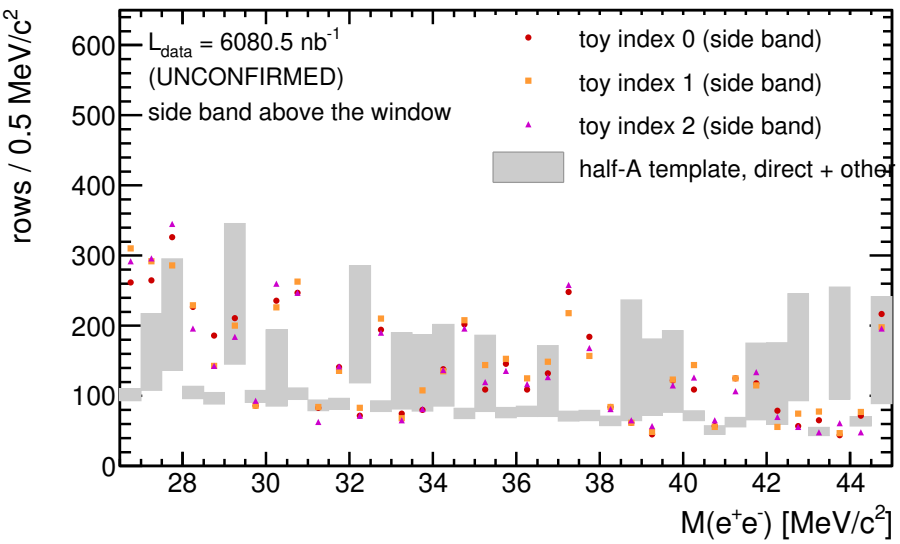


Figure 18: $M(e^+e^-)$ side band (above the X17 window) of three X17 toys against the half-A template (grey boxes). All toys show the same spikes, which come from single heavy rows of half B. $L_{\mathrm{data}} = 6080.5\,\mathrm{nb}^{-1}$ (UNCONFIRMED).

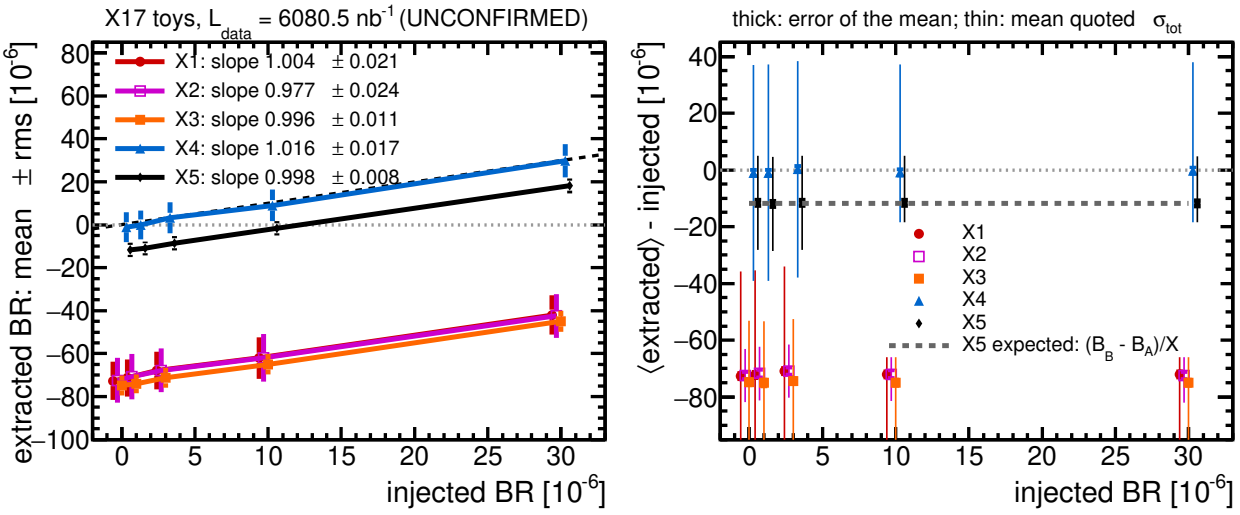


Figure 19: Blind test, X17. Left: extracted against injected BR (mean ± rms over 300 toys) with straight-line fits. Right: mean offset; thick bars: error of the mean, thin bars: mean quoted error. Dotted: the offset predicted for X5. $L_{\text{data}} = 6080.5\,\text{nb}^{-1}$ (UNCONFIRMED).

Limits of this test. The test is a closure test inside one model. It does not show how the methods behave in data.

- Templates and pseudo-data come from the same SMASH model; there is no CB and no data/SMASH shape difference (0.87–1.36 above $0.56\,\text{GeV}/c^2$ in earlier comparisons). Template fits are favoured by construction. L_{toy} is 55 times below L_{data}.
- **The test was not blind by construction.** The frozen tables that the analyst read contain full-sample (A+B) numbers; together with the half-A templates they give the half-B truth (for example the PEAK yield 241.4 and the X17 background of half B). No sign was found that these numbers were used, and the sealed key was not read before the results were written, but the leak was possible.
- The bootstrap repeats identical rows; how often a row recurs across toys reveals its set and therefore class 4 (std) against class 5 (η set). This was not exploited, but it is a second possible leak.
- The X17 part used the v1 selection, whose $M(\pi^+\pi^-e^+e^-)$ window was optimised (partly on a step of 4 to 8 std rows); the fixed window raises UL_{stat} by 3 %.

5.8 What this means for the data analysis

1. **Cuts.** In the simulation the four η-selection cuts remove background outside the windows but do not improve the windows themselves; OA$> 2°$ and $\alpha_\eta > 140°$ cost signal (conversions, bump). Their use in data must be justified by the CB and checked there. The pion-vertex cuts are the only ones with a gain inside the windows (4–6 %); the frozen PEAK and DIRECT cuts need the pion track z in the data compact rows, otherwise "pre + pions in vertex fit" is the alternative.
2. **Peak extraction.** Template fit (M2) and Gaussian + polynomial with calibration (M1) close within a few per cent in simulation; a difference of about 12 % between them is normal. Linear side bands under the peak lose 20–45 % and should not be used.
3. **Bump.** Side band with the simulated background shape (B4) and template fit; the parametric fit tends to be high.

4. **Direct decay alone.** The split of the peak into class 5 and class 4 is limited by the 307 std rows of class 4; the quoted error must include the simulation statistics.

5. **Simulation statistics.** The std set limits the bump and conversion templates (N_{eff} 270–1332) and the X17 background. The common MC error must be added to every yield.

6. **X17.** With the background taken from this simulation the expected limit is about 2×10^{-5} (at L_{data} UNCONFIRMED), set by the simulation statistics of the background; 5.7×10^{-6} would need the background known exactly. Any X17 estimate must carry the background uncertainty in its error. A larger std production or a background determined from data side bands would be needed to approach the statistical limit.

Open points of this section. (i) L_{data} is UNCONFIRMED. (ii) The absolute efficiencies rest on $N_{\text{direct}} = 100 \times 1\,181\,801$ (representativeness of the 1000 generator files UNVERIFIED). (iii) The X17 $M(\pi^+\pi^-)$ distribution and decay angle are assumed (phase space, isotropic). (iv) The blind test should be repeated with cut definitions only, with hidden per-toy signal scales and with pseudo-data that do not repeat identical rows; a version with a CB-like component and a reweighted background shape would test the methods closer to data. (v) The pion-vertex variables were not compared between simulation and data. (vi) The beam energy (4530 in the analysis, 4500 MeV in SMASH) does not affect $M(\pi^+\pi^-e^+e^-)$, $M(e^+e^-)$ and $M(\pi^+\pi^-)$, but does affect MM_4.

6 Next steps: the data

Section 5 covered the first four steps of the planned study (signal, acceptance, cut flow, backgrounds) and tested the optimisation and the signal extraction on simulation. The next steps use the data:

1. **Combinatorial background.** The like-sign method and the factor k from data; the signal after subtraction, checked in regions without the η. Only here can the cuts that the simulation finds useless inside the windows (OA, α_{lab}, α_η) show a gain, through the CB.

2. **Comparison with the simulation** in η-enriched selections below about $1\,\text{GeV}/c^2$, with the physics weights of Sect. 3.13.

3. **Signal extraction.** The bump and the peak with the methods that passed the blind test (template fit, Gaussian with polynomial; side band with the simulated background shape for the bump), with the spread between methods as a systematic error.

4. **Normalisation.** The ratio of the peak to the bump, in which the luminosity and much of the acceptance cancel, gives $\text{BR}(\eta \to \pi^+\pi^-e^+e^-)$ relative to $\text{BR}(\eta \to \pi^+\pi^-\pi^0)\,\text{BR}(\pi^0 \to e^+e^-\gamma)$.

5. **Systematic errors.** PID cut widths, the opening-angle cut and conversions, vertex cuts, the physics weights, the beam energy, the simulation statistics of the templates.

6. **X17.** The $M(e^+e^-)$ spectrum in the peak with the background constrained from data, and an upper limit.

A File locations

PAT outputs	`/hdd3/przygoda/HADES/files/sim/ppee_{std,lep,eta}_all.root`
FAT outputs	`/hdd3/przygoda/HADES/FAT_out/sim/pippimepem_betanew/output_<set>_<channel>.root`
Compact files	`.../pippimepem_betanew/scan/sim_compact.root`, `sim_compact_eta.root`
PID cuts	simulation `pp45_pid_cuts_sim_betanew_v6.root`, experiment `pp45_pid_cuts_exp_betanew_v1.root` (PAT repository)
PID samples (vae25)	`/lustre/hades/user/przygoda/pat3/out/sim/HADRONS/smash_std_pidsamples/HH_elossfix/`, `.../out/sim/LEPTONS/smash_std_pidsamples/LEP_merged/`
Report macros	`docs/latex/macros/`: `banana_fill.C` (run on vae25), `rep_banana.C`, `rep_stats.C`